Cardiovascular OSCE Cases

Question and answer review by exam year, 2015 to 2025

Key: Drug names **Critical point** ★ Repeated case Answers follow ACC/AHA, ESC, CHEST and ADA guidance.

2025

Case 1: HTN, HFrEF, AF, Hyperthyroidism

Case details: BP 150/90, EF 35%, symptomatic, ischemic heart disease 10 years ago. Home: Furosemide 40 mg OD, Lisinopril 5 mg OD, Atorvastatin 10 mg OD, Amiodarone 200 mg OD.

Question	Answer
Q1. Is there any modification for the statin? Why?	• Yes: switch to a **high-intensity statin** • Why: ischemic heart disease = clinical ASCVD, so high intensity is indicated (age ≤75) • Options: Atorvastatin 40–80 mg daily or Rosuvastatin 20–40 mg daily • Amiodarone interaction: caps only Simvastatin (max 20 mg) and Lovastatin (max 40 mg); Atorvastatin and Rosuvastatin need no cap • Goal: LDL-C reduction ≥50%; LDL-C <55 mg/dL is reasonable for secondary prevention • Next step: add Ezetimibe 10 mg if LDL-C stays ≥70 mg/dL on maximally tolerated statin • Recheck: lipid panel 4–12 weeks after the change
Q2. Management of HTN?	• Goal: BP <130/80 mm Hg • Lower BP with GDMT itself: ARNI, evidence-based beta-blocker, MRA and SGLT2i all lower BP and reduce mortality • Avoid: non-DHP CCBs (Diltiazem, Verapamil) in HFrEF (harm) • Lifestyle: sodium <2300 mg/day, weight loss, DASH diet, limit alcohol • Follow-up: BP, SCr and K 1–2 weeks after each change
Q3. Is Lisinopril needed?	• RAAS inhibition is needed (mortality benefit), but not as Lisinopril • ARNI is preferred over ACE inhibitor in symptomatic HFrEF (class 1); replace Lisinopril with Sacubitril/valsartan • **Never give an ACE inhibitor together with an ARNI** (angioedema)
Q4. Team wants Entresto: what will you do, and what dose?	• Agree: ARNI lowers CV death and HF hospitalization more than ACE inhibitor (PARADIGM-HF) • Washout: stop Lisinopril and wait **36 hours** before the first ARNI dose • Dose: Lisinopril 5 mg/day is a low dose (≤10 mg/day), so start Sacubitril/valsartan **24/26 mg BID** • Titration: double every 2–4 weeks to target 97/103 mg BID • Monitor: BP, SCr and K at 1–2 weeks, angioedema, fluid status (diuretic may need reduction) • Note: BNP rises with neprilysin inhibition; follow NT-proBNP instead
Q5. GDMT?	• ARNI: Sacubitril/valsartan 24/26 mg BID, target 97/103 mg BID • Beta-blocker (start when euvolemic): ◦ Bisoprolol 1.25 mg daily, target 10 mg daily ◦ Carvedilol 3.125 mg BID, target 25 mg BID (50 mg BID if >85 kg) ◦ Metoprolol succinate 12.5–25 mg daily, target 200 mg daily • MRA: Spironolactone 12.5–25 mg daily if eGFR >30 and K <5.0 • SGLT2i: Dapagliflozin 10 mg daily or Empagliflozin 10 mg daily • Loop diuretic: only for congestion; no mortality benefit • Device: ICD if EF stays ≤35% after ≥3 months of optimized GDMT • Monitor: BP, HR, SCr and K 1–2 weeks after each start/titration; weight and congestion each visit • Monitor MRA: K and SCr ~1 week, 1 month, 6 months; halve or stop if K >5.5 • Monitor SGLT2i: volume status, genital/urinary infection, euglycemic DKA • Non-pharm: sodium <2300 mg/day, daily weights, exercise/cardiac rehab, vaccines, avoid NSAIDs and non-DHP CCBs • Add-on options if symptoms persist despite optimized GDMT: Vericiguat 2.5 mg daily, doubled every 2 weeks to 10 mg (VICTORIA, recent worsening HF; avoid with PDE5 inhibitors); Ivabradine 5 mg BID (sinus rhythm only, resting HR ≥70 on maximal beta-blocker, so not in AF); Hydralazine + Isosorbide dinitrate (African American patients or ACE inhibitor/ARNI intolerance)
Q6. How to manage the signs and symptoms?	• Congestion: loop diuretic is the cornerstone • Outpatient: increase Furosemide (eg, 40 mg BID) and reassess weight in days • If admitted: IV Furosemide (20 mg IV = 40 mg PO); double the dose until response, eg urine output >500 mL in 2 hours • Refractory: add Metolazone 2.5–5 mg PO 30 min before the loop, or Chlorothiazide 250–500 mg IV • Goals: weight loss, less edema and dyspnea, euvolemia on the lowest diuretic dose • Supportive: daily weights, sodium restriction, fluid restriction only if hyponatremia or advanced HF • Do not up-titrate the beta-blocker while the patient is fluid overloaded
Q7. On screen: hyperthyroidism. Interpretation?	• Labs: low TSH with high free T4 (and/or T3) = thyrotoxicosis • In this patient: Amiodarone-induced thyrotoxicosis (AIT)

Question	Answer
Q8. Why will you treat?	• Thyrotoxicosis worsens HFrEF: tachycardia, higher myocardial oxygen demand, decompensation • Arrhythmia: precipitates AF recurrence and ventricular arrhythmias • Outcome: AIT in LV dysfunction carries higher mortality, so treat promptly
Q9. How will you treat?	• First, define the type (thyroid Doppler, radioiodine uptake, history of thyroid disease): ◦ Type 1: iodine load drives excess hormone synthesis in an abnormal gland → thionamide ◦ Type 2: destructive thyroiditis → corticosteroid ◦ Mixed or unclear: thionamide plus corticosteroid • Symptoms: beta-blocker (already part of GDMT) • Amiodarone: decide with cardiology; may be continued if needed for life-threatening arrhythmia; stopping has little short-term effect (half-life ~58 days) • Monitor: free T4/T3 every 4–6 weeks, CBC and LFTs on thionamide
Q10. Dose?	• Type 1: Methimazole 40 mg daily, taper as thyroid function normalizes • Type 2: Prednisone 40 mg daily for 2–4 weeks, then taper over 2–3 months • Mixed: both regimens together
Q11. What caused his hyperthyroidism?	• Amiodarone: about 37% iodine by weight • Type 1 mechanism: iodine-induced excess hormone synthesis • Type 2 mechanism: direct toxic thyroiditis releasing preformed hormone
Also be ready for (exam add-ons)	• Goals: reduce mortality and HF hospitalization, relieve congestion and symptoms, prevent remodeling, improve exercise tolerance and quality of life • Non-pharm: sodium <2–3 g/day, fluid 1.5–2 L only if hyponatremia or advanced HF, daily weights, exercise or cardiac rehabilitation, vaccines (influenza, pneumococcal, COVID-19), stop smoking, limit alcohol, avoid NSAIDs • First-line: 4 pillars: ARNI Sacubitril/valsartan 24/26 mg BID (or ACE inhibitor/ARB), Bisoprolol 1.25 mg (or Carvedilol 3.125 mg BID, Metoprolol succinate 12.5–25 mg) when euvolemic, Spironolactone 12.5–25 mg (eGFR >30, K <5), Dapagliflozin 10 mg or Empagliflozin 10 mg; Furosemide 20–40 mg only for congestion • Pregnancy: ACE inhibitor, ARB, ARNI, MRA and SGLT2 inhibitors are contraindicated; use Metoprolol, Labetalol, Hydralazine + Isosorbide dinitrate and a loop diuretic; peripartum cardiomyopathy: add Bromocriptine and anticoagulation per specialist; counsel contraception • Renal/dialysis: check SCr and K (ARNI, ACE inhibitor, MRA); avoid MRA if eGFR <30 or K >5; Empagliflozin/Dapagliflozin can start at eGFR ≥20; higher loop-diuretic doses needed; Digoxin reduce in CKD • Liver: avoid ARNI in Child-Pugh C; Carvedilol avoid in severe disease; congestive hepatopathy improves with decongestion • GI (PUD/GERD): if on antiplatelet or anticoagulant add a PPI (Pantoprazole preferred with Clopidogrel); avoid NSAIDs • Counseling: ARNI: dizziness, angioedema, hyperkalemia, 36-hour ACE-inhibitor washout; β-blocker: fatigue, do not stop abruptly; Spironolactone: avoid potassium salt substitutes, gynecomastia; SGLT2 inhibitor: genital infection, hydration, hold on sick days; weigh daily and call if gain >1 kg in a day or 2 kg in a week • Hyperthyroidism (side problem): Methimazole 10–30 mg daily (Propylthiouracil in first trimester); Propranolol can control rate; Amiodarone-induced thyrotoxicosis needs specialist care • Pregnancy (thyroid): Propylthiouracil in the first trimester then switch to Methimazole; avoid Amiodarone (fetal goiter and bradycardia)

Case 2: Pulmonary Embolism (PCOS)

Case details: Young woman, known PCOS on hormonal treatment, developed massive PE and DVT with clear signs, symptoms and objective data. Weight 62.4 kg.

Question	Answer
Q1. Subjective and objective data for DVT and PE?	• DVT subjective: unilateral leg pain, swelling, heaviness • DVT objective: calf asymmetry, warmth, erythema, positive compression ultrasound • PE subjective: sudden dyspnea, pleuritic chest pain, cough, hemoptysis, syncope • PE objective: tachycardia, tachypnea, hypoxemia, CT pulmonary angiography filling defect, RV dysfunction on echo, raised troponin/BNP, elevated D-dimer • Massive PE marker: **SBP <90 mm Hg or shock** • Risk factor: estrogen-containing hormonal therapy
Q2. They will start Enoxaparin: what dose?	• Dose: Enoxaparin 1 mg/kg SC q12h = 62.4 mg, round to **60 mg SC q12h** • Alternative: 1.5 mg/kg SC daily (~90 mg) • Check: CrCl (q24h if <30 mL/min), CBC • Note: in massive PE where lysis is planned, IV UFH 80 units/kg then 18 units/kg/h is preferred (short half-life, reversible)

Question	Answer
Q3. They will use a fibrinolytic: what is the rationale?	• Indication: massive (high-risk) PE with hypotension and no high bleeding risk • Benefit: dissolves clot rapidly, relieves RV pressure overload, restores perfusion, lowers death and hemodynamic collapse • Regimen: Alteplase 100 mg IV over 2 hours • Heparin: hold during the infusion; resume when aPTT <2× control • Screen first: absolute contraindications (prior ICH, recent stroke, active bleeding, recent major surgery) • Monitor: BP and neuro checks every 15 min during and after infusion, bleeding signs, Hgb, aPTT before restarting UFH, RV function/O2 response
Q4. Warfarin started; after 6 days INR 2.3. What will you do?	• Bridging rule: overlap parenteral with Warfarin **≥5 days AND until INR ≥2.0 for 24 hours** • Action: day 6 with INR 2.3, confirm INR ≥2 for 24 h, then stop Enoxaparin • Warfarin: continue the same dose; target INR 2–3 • Also: stop estrogen-containing therapy; counsel on non-estrogen contraception (Warfarin is teratogenic)
Q5. She came with INR 3.7. What will you do, and when to follow up?	• Assess: bleeding (none assumed), cause (interacting drug, diet, illness, missed/extra dose) • Action: hold or skip one dose, then reduce the total weekly dose by 5–20% • If single value ≤0.5 above range in a stable patient: may continue the same dose • Follow-up: recheck INR in 3–7 days, then weekly until stable
Q6. She developed unprovoked seizures and they want to start Carbamazepine. What will you do?	• Interaction: Carbamazepine is a strong CYP3A4, 2C9 and 1A2 inducer → **INR falls**, recurrent VTE risk; effect builds over 1–4 weeks and reverses slowly when stopped • Recommend a non-inducing agent: Levetiracetam 500 mg BID (or Lacosamide, Lamotrigine) • If Carbamazepine is still chosen: expect a large Warfarin dose increase; check INR weekly (or twice weekly early) until stable
Q7. INR came 1.3.	• Interpretation: subtherapeutic, likely Carbamazepine induction (also check adherence, diet) • Bridge: recent PE = high recurrence risk → restart Enoxaparin 1 mg/kg SC q12h until INR ≥2 • Warfarin: increase the total weekly dose by 10–20% (larger increases often needed with an inducer) • Recheck: INR in 3–7 days
Q8. INR came 1.8.	• Interpretation: slightly below range (within 0.5 of target) • Action: increase the weekly dose by 5–10% because the inducer is ongoing • Bridging: not routinely needed at this level • Recheck: INR within 1 week
Also be ready for (exam add-ons)	• Goals: stop clot propagation, prevent death and recurrence, restore hemodynamics (if massive), prevent post-thrombotic syndrome and CTEPH • Non-pharm: early ambulation, compression stockings only for symptoms, hydration, stop smoking, weight control, review hormonal therapy (stop estrogen-containing contraceptives) • First-line: Enoxaparin 1 mg/kg q12h (or 1.5 mg/kg daily) or UFH 80 units/kg bolus then 18 units/kg/h (massive PE, planned fibrinolysis); DOAC: Apixaban 10 mg BID × 7 days then 5 mg BID, Rivaroxaban 15 mg BID × 21 days then 20 mg daily; massive PE: Alteplase 100 mg over 2 h; Warfarin INR 2–3 • Pregnancy: LMWH is the drug of choice (weight-based, no routine anti-Xa), UFH if delivery or fibrinolysis near; Warfarin and DOACs contraindicated; continue until 6 weeks postpartum (minimum 3 months total); Warfarin and Enoxaparin are safe during breastfeeding • Renal/dialysis: Enoxaparin 1 mg/kg q24h if CrCl <30 (anti-Xa monitoring); prefer UFH in dialysis; Apixaban needs no adjustment for VTE; Dabigatran, Edoxaban and Rivaroxaban avoided in severe CKD • Liver: avoid DOACs in Child-Pugh C; Warfarin has a high baseline INR; LMWH acceptable • HF/cancer: LMWH or Apixaban/Rivaroxaban/Edoxaban in cancer; avoid Warfarin if possible; Dabigatran with GI cancers carries more bleeding • Interactions: Carbamazepine, Phenytoin, Rifampin induce Warfarin and DOAC metabolism (use LMWH or an alternative anticonvulsant); estrogen-containing contraceptives are contraindicated after VTE (use IUD or progestin) • Counseling: injection technique and site rotation, bleeding precautions, avoid NSAIDs/Aspirin, signs of recurrence, duration (≥3 months; indefinite if unprovoked or recurrent), Warfarin diet and INR schedule

Case 3: STEMI ★ Repeated case, also in 2024, 2022–2023, 2018–2019

Case details: Typical STEMI. Home: Atorvastatin only. Labs and vitals normal except troponin. LDL-C 80 mg/dL.

Question	Answer
Q1. What will you give in the ER, with doses and route?	• Aspirin: 162–325 mg PO, chewed, non-enteric coated • P2Y12 load: Ticagrelor 180 mg PO or Prasugrel 60 mg PO (preferred); Clopidogrel 600 mg if neither possible • Anticoagulant for primary PCI: UFH 70–100 units/kg IV bolus (50–70 units/kg if GP IIb/IIIa used), titrated to ACT; or Bivalirudin 0.75 mg/kg IV then 1.75 mg/kg/h • Nitroglycerin: 0.4 mg SL q5min × 3 for ongoing pain; IV 10 mcg/min if persistent • Oxygen: only if SaO2 <90% or respiratory distress • Morphine: 2–4 mg IV only if pain persists despite anti-ischemic therapy • Beta-blocker: oral within 24 h if no HF, low output or shock • High-intensity statin: Atorvastatin 80 mg PO • Monitor: continuous ECG, BP, HR, SaO2, pain, troponin, Hgb/platelets, SCr, ACT during PCI

Question	Answer
Q2. What is the best management strategy?	• Primary PCI within symptom onset ≤12 h • Time goal: first medical contact to device ≤90 min (≤120 min if transferred) • Fibrinolysis only if PCI cannot be done within 120 min
Q3. What are the benefits of it?	• Restores patency of the infarct-related artery • Limits infarct size and preserves LV function • Reduces mortality, reinfarction and HF
Q4. Benefits of PCI over fibrinolysis?	• Higher patency: TIMI 3 flow >90% vs about 50–60% with lysis • Lower: mortality, reinfarction, stroke and intracranial hemorrhage • Defines anatomy and treats the culprit lesion in one step • No lytic contraindication issue (recent surgery, bleeding, prior ICH)
Q5. Long-term medications (classes only)?	• DAPT: Aspirin + P2Y12 inhibitor ≥12 months, then single antiplatelet • High-intensity statin ± Ezetimibe ± PCSK9 inhibitor • Beta-blocker • ACE inhibitor/ARB (EF ≤40%, HTN, DM, CKD) • MRA if EF ≤40% with HF or DM • SL Nitroglycerin PRN • PPI if high GI bleeding risk • Cardiac rehabilitation (class 1, often missed)
Q6. Do you want to adjust his DLP regimen?	• Yes: ACS on a statin with LDL-C 80 mg/dL is above goal • Increase: Atorvastatin to 80 mg daily • Add now: Ezetimibe 10 mg daily (already on statin with LDL-C ≥70) • Goal: LDL-C <55 mg/dL is reasonable after ACS • Recheck: lipids in 4–8 weeks; add PCSK9 mAb (Evolocumab, Alirocumab) or Inclisiran if still above goal
Q7. He developed AKI. Which medication will you adjust?	• No change: Aspirin, Ticagrelor/Prasugrel, Atorvastatin, beta-blocker • Hold until recovery: ACE inhibitor/ARB, MRA, NSAIDs, Metformin • Renally adjust if used: ◦ Enoxaparin q24h if CrCl <30 ◦ Eptifibatide infusion 50% if CrCl <50; Tirofiban infusion 50% if CrCl ≤60 ◦ Rosuvastatin max 10 mg if CrCl <30 (Atorvastatin needs no change) • Avoid further contrast and nephrotoxins
Also be ready for (exam add-ons)	• Goals: restore infarct-artery patency quickly (door-to-balloon ≤90 min, door-to-needle ≤30 min), limit infarct size, relieve ischemia, prevent arrhythmia, heart failure and reinfarction • Non-pharm: primary PCI (preferred), cardiac rehab, smoking cessation, Mediterranean diet, exercise 150 min/week, weight control, sodium restriction, vaccines (influenza), driving advice • First-line: Aspirin 162–325 mg chewed, P2Y12 inhibitor (Ticagrelor 180 mg or Prasugrel 60 mg or Clopidogrel 600 mg), UFH 70–100 units/kg in the cath lab, Atorvastatin 80 mg, oral β-blocker within 24 h if LVEF reduced, HF or ventricular arrhythmia and no shock (2025 ACC/AHA; reassess after 1 year if LVEF is preserved), Nitroglycerin SL for pain • Pregnancy: MI in pregnancy is rare: PCI preferred, avoid fibrinolysis if possible; low-dose Aspirin and Clopidogrel acceptable; avoid ACE inhibitors/ARBs/MRA and statins; Metoprolol and Labetalol are safe; UFH or Enoxaparin acceptable • Renal/dialysis: contrast nephropathy prevention (hydration, minimal contrast, hold Metformin and nephrotoxins); Enoxaparin 1 mg/kg q24h if CrCl <30; Bivalirudin: reduce the infusion if CrCl <30; Ticagrelor needs no adjustment • Liver: avoid Ticagrelor in severe hepatic impairment; statin caution when ALT >3× ULN; bleeding risk higher • HF/shock: avoid β-blocker in acute HF or shock; add Spironolactone if EF <40% with DM or HF; ACE inhibitor/ARNI for EF <40% • GI (PUD/GERD): add PPI with DAPT in patients at GI-bleeding risk (Pantoprazole; avoid Omeprazole with Clopidogrel) • LDL-C goal <55 mg/dL (add Ezetimibe if ≥70 on maximal statin, 2025 ACC/AHA); Counseling: DAPT adherence for 12 months (never stop early), bleeding signs, Ticagrelor dyspnea and Aspirin dose ≤100 mg, Nitroglycerin use (call 911 if pain persists 5 min after first dose), statin muscle pain, cardiac rehab, return to activity

Case 4: Stroke + AF

Case details: Patient with AF who presented with ischemic stroke. On Amiodarone, Warfarin, Fluoxetine and Insulin during follow-up.

Question	Answer
Q1. Calculate CHA2DS2-VASc and HAS-BLED for this patient.	• CHA2DS2-VASc: CHF 1, HTN 1, Age ≥75 2, DM 1, Stroke/TIA/embolism 2, Vascular disease 1, Age 65–74 1, Female 1 • This patient: prior stroke alone = 2 points, so score is **≥2** • HAS-BLED: HTN (SBP >160) 1, Abnormal renal 1 and liver 1, Stroke 1, Bleeding 1, Labile INR 1, Elderly >65 1, Drugs 1 and alcohol 1 • This patient: stroke = 1 point at least

Question	Answer
Q2. Interpretation of the scores?	• CHA2DS2-VASc ≥2 (men) or ≥3 (women): oral anticoagulation recommended • HAS-BLED ≥3: high bleeding risk; **not a reason to withhold OAC**; fix modifiable risks (BP, NSAIDs, alcohol, labile INR) and follow closely
Q3. Would you start anticoagulants?	• Yes: secondary prevention after cardioembolic stroke • Preferred: DOAC over Warfarin (unless mechanical valve or moderate–severe mitral stenosis) • Timing by stroke severity: minor/moderate within ~48 h; major around day 6–7, after imaging excludes hemorrhagic transformation • Dose: Apixaban 5 mg BID (or Rivaroxaban 20 mg daily with evening meal, 15 mg if CrCl 15–50) • Monitor: bleeding, Hgb, SCr and LFTs at baseline and yearly, adherence, new neuro deficits
Q4. What is the target INR?	• INR 2.0–3.0 (target 2.5)
Q5. Effect of Amiodarone on Warfarin? What do you do with the Warfarin dose?	• Mechanism: Amiodarone inhibits CYP2C9 (S-Warfarin), 1A2 and 3A4 → **INR rises** • Time course: within 1–2 weeks, maximal over weeks; persists for months after stopping • Action: reduce the Warfarin dose empirically by 30–50% • Monitor: INR weekly until stable
Q6. A few days later: simple nasal bleeding with INR 4.7. What will you do?	• Minor bleed, INR 4.5–10: ◦ Local measures (pressure, topical vasoconstrictor) ◦ Hold Warfarin ◦ May give low-dose oral Vitamin K 2.5 mg ◦ Check INR daily • Not IV Vitamin K 10 mg: that is for major or life-threatening bleeding (with 4F-PCC) • Find the cause: Amiodarone, Fluoxetine, diet, illness
Q7. New Warfarin dose, and when will you monitor?	• Restart when INR is back in range • Dose: reduce the total weekly dose by 10–20% • Monitor: INR within 3–7 days (a dose change takes 3–5 days to show), then weekly until stable
Q8. DDIs between fluoxetine and Amiodarone, insulin and Warfarin, and what to do?	• Fluoxetine + Amiodarone: additive **QT prolongation** (torsades) ◦ ECG/QTc at baseline and after; keep K >4 and Mg >2; consider Sertraline instead • Fluoxetine + Insulin: SSRIs increase insulin sensitivity → hypoglycemia ◦ More frequent glucose checks; insulin dose may need reduction • Fluoxetine + Warfarin: CYP2C9 inhibition raises INR, plus SSRI platelet effect raises bleeding ◦ INR within 1 week of starting or stopping; counsel on bleeding signs
Q9. After 1 year without follow-up, he came with stroke. Anticoagulants? Is there a need for antiplatelets?	• Find why: adherence, INR at presentation, time in therapeutic range • Anticoagulation: restart/optimize OAC once safe; switch to a DOAC (fixed dose, no INR) given poor follow-up • If stroke occurred on therapeutic OAC: look for another cause; do not switch or add drugs empirically • Antiplatelet: No. Adding Aspirin to OAC raises bleeding without lowering stroke, unless a separate indication (recent ACS or stent) • Acute window: thrombolysis only if INR ≤1.7 and no DOAC in the last 48 h • Monitor: adherence at each visit, INR (if Warfarin) weekly until stable, bleeding, BP control
Also be ready for (exam add-ons)	• Goals: prevent stroke (anticoagulation), control rate or rhythm, treat the cause, improve symptoms and prevent HF/tachycardiomyopathy • Non-pharm: weight loss, alcohol reduction, treat sleep apnea, exercise, BP control, smoking cessation, avoid stimulants (caffeine, decongestants); catheter ablation if drug failure • First-line: anticoagulant by CHA2DS2-VASc (men ≥2, women ≥3): Apixaban 5 mg BID (2.5 mg if 2 of age ≥80, weight ≤60 kg, SCr ≥1.5), Rivaroxaban 20 mg with the evening meal (15 mg if CrCl 15–50), or Warfarin INR 2–3 (valvular AF); rate control: Metoprolol/Bisoprolol or Diltiazem (not in HFrEF); rhythm: Flecainide/Propafenone (no structural disease), Amiodarone/Dofetilide (HF, CAD) • Pregnancy: DOACs and Warfarin contraindicated; use LMWH (Enoxaparin 1 mg/kg q12h) or UFH; rate control with Metoprolol or Labetalol; cardioversion is safe; avoid Amiodarone, Dronedarone • Renal/dialysis: Apixaban 5 mg BID (2.5 mg if age ≥80 or weight ≤60) in HD; Dabigatran avoid if CrCl <30; Rivaroxaban 15 mg if CrCl 15–50, avoid <15; Edoxaban 30 mg if CrCl 15–50; Dofetilide/Sotalol adjust to CrCl • Liver: avoid DOACs in Child-Pugh C (Rivaroxaban avoided in B and C); Amiodarone monitor LFTs; Warfarin baseline INR is high • HF: avoid Flecainide, Propafenone, Dronedarone, Sotalol and Diltiazem/Verapamil in HFrEF; use Amiodarone or Dofetilide; β-blocker for rate • GI (PUD/GERD): PPI when combined with antiplatelets; bleeding risk; HAS-BLED corrections (stop NSAIDs, control BP) • Counseling: adherence (DOAC half-life is short), bleeding signs, Rivaroxaban with food, Warfarin: steady Vitamin K intake and INR checks, drug interactions (Amiodarone, Clarithromycin, Rifampin, Carbamazepine), missed dose rules, show the card to all clinicians • Stroke + AF: start the anticoagulant early: within 48 h after a minor or moderate stroke and from day 6–7 after a major stroke (ELAN); the older 1-3-6-12 rule is more conservative; no antiplatelet on top unless a separate indication; Amiodarone lowers Warfarin dose by 30–50%; Fluoxetine inhibits CYP2C9 (raise INR)

2024

Case 1: Atrial Fibrillation then STEMI ★ Repeated case, also in 2022–2023, 2021, 2018–2019

Case details: 69-year-old man with AF, HTN, DM on Warfarin, Metformin, Lisinopril, Flecainide, Diltiazem. HR 99 with fatigue and symptoms. Developed STEMI and received PCI.

Question	Answer
Q1. Risk factors for stroke in this patient?	• Non-modifiable: age 69, male sex • Modifiable: AF, HTN, DM • After STEMI: vascular disease (MI)
Q2. Calculate CHA2DS2-VASc.	• HTN 1 + DM 1 + Age 65–74 1 + Vascular (MI) 1 = **4** • Interpretation: high risk; oral anticoagulation indicated
Q3. Calculate HAS-BLED (missing information in the case).	• Elderly >65: 1 • Drugs (antiplatelet after PCI): 1 • Score: 2 (renal/liver function, bleeding history, labile INR and SBP >160 not given)
Q4. He developed STEMI: what medication changes will you make?	• Stop Flecainide: class Ic is contraindicated in CAD/structural heart disease (CAST: higher mortality) ◦ Rhythm alternatives: Amiodarone, Dofetilide, Sotalol (if EF preserved) • Replace Diltiazem with a beta-blocker: Metoprolol succinate or Bisoprolol (post-MI benefit) • Antithrombotic: switch Warfarin to Apixaban + Clopidogrel 75 mg; Aspirin only peri-PCI up to 1 week (max 30 days) • Statin: Atorvastatin 40–80 mg daily • Continue Lisinopril (DM, HTN); hold Metformin 48 h after contrast • Regimen with doses: ◦ Apixaban 5 mg BID (2.5 mg if ≥2 of age ≥80, weight ≤60 kg, SCr ≥1.5) ◦ Clopidogrel 75 mg daily for 12 months, then OAC alone ◦ Aspirin 81 mg only until discharge or ≤1 week (max 30 days) ◦ Metoprolol succinate 25–50 mg daily, titrate to HR ◦ Atorvastatin 80 mg daily ◦ Rhythm option: Amiodarone 400 mg BID–TID to ~6–10 g, then 200 mg daily • Monitor: bleeding and Hgb, SCr (DOAC dosing) at least yearly, HR/BP, ECG (QTc if antiarrhythmic), lipids 4–8 weeks
Q5. Shifted from Warfarin to Apixaban: criteria for reducing the dose?	• 2.5 mg BID if ≥2 of: age ≥80 years, weight ≤60 kg, SCr ≥1.5 mg/dL • Also reduce with strong dual CYP3A4 and P-gp inhibitors • This patient: none → Apixaban 5 mg BID
Q6. How will you shift from Warfarin to Apixaban?	• Stop Warfarin, start Apixaban when INR <2.0
Q7. Target HR and your management?	• Target: lenient resting HR <100–110; **<80 at rest** if still symptomatic or with HF/LV dysfunction • This patient (symptomatic at 99): aim <80 • Management: stop Diltiazem, start Bisoprolol 2.5–5 mg daily (or Metoprolol succinate) and titrate to HR • Monitor: resting HR and BP at each visit, symptoms, ECG, fatigue/dizziness
Also be ready for (exam add-ons)	• Goals: prevent stroke (anticoagulation), control rate or rhythm, treat the cause, improve symptoms and prevent HF/tachycardiomyopathy • Non-pharm: weight loss, alcohol reduction, treat sleep apnea, exercise, BP control, smoking cessation, avoid stimulants (caffeine, decongestants); catheter ablation if drug failure • First-line: anticoagulant by CHA2DS2-VASc (men ≥2, women ≥3): Apixaban 5 mg BID (2.5 mg if 2 of age ≥80, weight ≤60 kg, SCr ≥1.5), Rivaroxaban 20 mg with the evening meal (15 mg if CrCl 15–50), or Warfarin INR 2–3 (valvular AF); rate control: Metoprolol/Bisoprolol or Diltiazem (not in HFrEF); rhythm: Flecainide/Propafenone (no structural disease), Amiodarone/Dofetilide (HF, CAD) • Pregnancy: DOACs and Warfarin contraindicated; use LMWH (Enoxaparin 1 mg/kg q12h) or UFH; rate control with Metoprolol or Labetalol; cardioversion is safe; avoid Amiodarone, Dronedarone • Renal/dialysis: Apixaban 5 mg BID (2.5 mg if age ≥80 or weight ≤60) in HD; Dabigatran avoid if CrCl <30; Rivaroxaban 15 mg if CrCl 15–50, avoid <15; Edoxaban 30 mg if CrCl 15–50; Dofetilide/Sotalol adjust to CrCl • Liver: avoid DOACs in Child-Pugh C (Rivaroxaban avoided in B and C); Amiodarone monitor LFTs; Warfarin baseline INR is high • HF: avoid Flecainide, Propafenone, Dronedarone, Sotalol and Diltiazem/Verapamil in HFrEF; use Amiodarone or Dofetilide; β-blocker for rate • GI (PUD/GERD): PPI when combined with antiplatelets; bleeding risk; HAS-BLED corrections (stop NSAIDs, control BP) • Counseling: adherence (DOAC half-life is short), bleeding signs, Rivaroxaban with food, Warfarin: steady Vitamin K intake and INR checks, drug interactions (Amiodarone, Clarithromycin, Rifampin, Carbamazepine), missed dose rules, show the card to all clinicians • Side scenario (post-MI): stop Flecainide/Propafenone (CAST) and Diltiazem if EF reduced; use a β-blocker, Amiodarone or Dofetilide; with PCI use Apixaban + Clopidogrel (stop Aspirin after 1 week to 1 month), then anticoagulant alone at 12 months; high-intensity statin

Case 2: HFrEF, AF, MI

Case details: HFrEF (EF ≤40%) with LVH, HF symptoms and arrhythmia; developed MI.

Question	Answer
Q1. What is the stage?	• Stage C: structural heart disease with current or prior HF symptoms
Q2. What is the management?	• Non-pharmacological: ◦ Sodium: avoid excess (<2300 mg/day) ◦ Fluid restriction 1.5–2 L/day only in advanced HF or hyponatremia ◦ Daily weights; report gain >1 kg/day or 2 kg/week ◦ Vaccines: influenza yearly, pneumococcal (PCV20 or PCV21), COVID-19 ◦ Cardiac rehabilitation, exercise, smoking cessation, limit alcohol ◦ Control BP <130/80, glucose and lipids ◦ Avoid NSAIDs, TZDs, non-DHP CCBs • GDMT (start low, titrate): ◦ ARNI Sacubitril/valsartan 24/26 mg BID → 97/103 mg BID (preferred) ◦ or ACE inhibitor Lisinopril 2.5–5 mg → 20 mg daily; or ARB Candesartan 4–8 mg → 32 mg daily ◦ Beta-blocker Bisoprolol 1.25 mg → 10 mg daily, Carvedilol 3.125 mg BID → 25 mg BID, or Metoprolol succinate 12.5–25 mg → 200 mg daily ◦ MRA Spironolactone 12.5–25 mg daily or Eplerenone 25 mg → 50 mg daily (eGFR >30, K <5.0) ◦ SGLT2i Dapagliflozin or Empagliflozin 10 mg daily ◦ Loop diuretic Furosemide 20–40 mg daily–BID for congestion • Post-MI: DAPT (or OAC + Clopidogrel if AF), high-intensity statin
Q3. What will you monitor and when?	• Efficacy: symptoms/NYHA class, weight, edema, BP, HR, titration to target doses • ACE inhibitor/ARB/ARNI: SCr and K 1–2 weeks after start or dose change (SCr rise up to 30% acceptable); BP, angioedema • MRA: K and SCr at about **1 week, 1 month and 6 months**, then periodically; halve or stop if K >5.5 • Beta-blocker: HR, BP, fluid status at 1–2 weeks • SGLT2i: SCr, volume status, genital/urinary infection, euglycemic DKA • Loop diuretic: weight, UOP, K, Mg, Na, SCr • Echo: repeat 3–6 months after GDMT optimization; ICD decision ≥40 days post-MI (≥90 days after revascularization)
Q4. What to look for in the ECHO?	• LVEF • LV size and dilation • LVH (wall thickness) • Regional wall motion abnormality (post-MI) • Valve disease (eg, mitral regurgitation) • Diastolic function and filling pressures • RV function and pulmonary pressure • LV thrombus
Q5. How will you treat AF with MI and HF?	• Rate control: evidence-based beta-blocker ± Digoxin; **avoid non-DHP CCB** • Rhythm control: Amiodarone or Dofetilide • Avoid: class Ic (Flecainide, Propafenone), Dronedarone with NYHA III–IV or recent decompensation • Anticoagulation: CHA2DS2-VASc (CHF + vascular already ≥2) → DOAC + Clopidogrel after PCI; Aspirin ≤1 week • Consider catheter ablation in HFrEF • Doses: ◦ Metoprolol succinate 25 → 200 mg daily or Carvedilol 3.125 → 25 mg BID ◦ Digoxin 0.125 mg daily (level 0.5–0.9 ng/mL) ◦ Amiodarone IV 150 mg over 10 min then 1 mg/min × 6 h, 0.5 mg/min × 18 h; oral 400 mg BID–TID to 6–10 g, then 200 mg daily ◦ Dofetilide 500 mcg BID if CrCl >60 (in hospital, QTc monitoring × 3 days) ◦ Apixaban 5 mg BID (2.5 mg if ≥2 of age ≥80, weight ≤60 kg, SCr ≥1.5) ◦ Clopidogrel 75 mg daily for 12 months, then OAC alone • Monitor: HR, ECG/QTc, Digoxin level, K/Mg, TFT/LFT (Amiodarone), bleeding, SCr

Question	Answer
Q6. Lifelong medications after MI?	• Antiplatelet: Aspirin 81 mg indefinitely after DAPT; with AF, OAC alone after 12 months • Beta-blocker: long term (EF ≤40%) • High-intensity statin ± Ezetimibe; LDL-C <55–70 mg/dL • ARNI/ACE inhibitor/ARB, MRA, SGLT2i (HFrEF) • SL Nitroglycerin PRN • Doses: ◦ Aspirin 81 mg daily (indefinitely) ◦ Ticagrelor 90 mg BID or Clopidogrel 75 mg daily for 12 months ◦ Atorvastatin 80 mg daily ± Ezetimibe 10 mg ◦ Metoprolol succinate 25 → 200 mg daily or Bisoprolol 1.25 → 10 mg daily ◦ Lisinopril 2.5–5 → 20–40 mg daily ◦ Nitroglycerin 0.4 mg SL q5min × 3 PRN chest pain • Monitor: chest pain, bleeding (Hgb), BP, HR, SCr/K (ACE inhibitor), lipids 4–8 weeks, LFT baseline, adherence to DAPT
Also be ready for (exam add-ons)	• Goals: reduce mortality and HF hospitalization, relieve congestion and symptoms, prevent remodeling, improve exercise tolerance and quality of life • Non-pharm: sodium <2–3 g/day, fluid 1.5–2 L only if hyponatremia or advanced HF, daily weights, exercise or cardiac rehabilitation, vaccines (influenza, pneumococcal, COVID-19), stop smoking, limit alcohol, avoid NSAIDs • First-line: 4 pillars: ARNI Sacubitril/valsartan 24/26 mg BID (or ACE inhibitor/ARB), Bisoprolol 1.25 mg (or Carvedilol 3.125 mg BID, Metoprolol succinate 12.5–25 mg) when euvolemic, Spironolactone 12.5–25 mg (eGFR >30, K <5), Dapagliflozin 10 mg or Empagliflozin 10 mg; Furosemide 20–40 mg only for congestion • Pregnancy: ACE inhibitor, ARB, ARNI, MRA and SGLT2 inhibitors are contraindicated; use Metoprolol, Labetalol, Hydralazine + Isosorbide dinitrate and a loop diuretic; peripartum cardiomyopathy: add Bromocriptine and anticoagulation per specialist; counsel contraception • Renal/dialysis: check SCr and K (ARNI, ACE inhibitor, MRA); avoid MRA if eGFR <30 or K >5; Empagliflozin/Dapagliflozin can start at eGFR ≥20; higher loop-diuretic doses needed; Digoxin reduce in CKD • Liver: avoid ARNI in Child-Pugh C; Carvedilol avoid in severe disease; congestive hepatopathy improves with decongestion • GI (PUD/GERD): if on antiplatelet or anticoagulant add a PPI (Pantoprazole preferred with Clopidogrel); avoid NSAIDs • Counseling: ARNI: dizziness, angioedema, hyperkalemia, 36-hour ACE-inhibitor washout; β-blocker: fatigue, do not stop abruptly; Spironolactone: avoid potassium salt substitutes, gynecomastia; SGLT2 inhibitor: genital infection, hydration, hold on sick days; weigh daily and call if gain >1 kg in a day or 2 kg in a week

Case 3: HFrEF ★ Repeated case, also in 2020

Case details: HTN on Amlodipine. HF signs and symptoms with minimal limitation of daily activity, no symptoms at rest. ECHO: EF <30%.

Question	Answer
Q1. What is the stage of HF, and why?	• Stage C: structural heart disease (EF <30%) with HF symptoms • NYHA class II: slight limitation; ordinary activity causes symptoms; comfortable at rest
Q2. What is the goal of treatment?	• Relieve signs and symptoms; improve exercise tolerance and quality of life • Reduce mortality • Reduce hospitalization and exacerbations • Prevent or reverse cardiac remodeling • Control risk factors and sodium/water retention
Q3. What is the evidence of structural heart disease from the ECHO?	• Reduced LVEF (<30%) • LV dilation • LVH • Wall motion abnormalities • Valvular disease
Q4. What is the management?	• Four GDMT pillars: ARNI, evidence-based beta-blocker, MRA, SGLT2i • Loop diuretic for congestion • Non-pharmacological as in Case 2 • Stop Amlodipine to make room for GDMT (see below) • Doses: ◦ Sacubitril/valsartan 24/26 mg BID (49/51 if on standard-dose ACE inhibitor/ARB) → 97/103 mg BID ◦ Bisoprolol 1.25 → 10 mg daily, Carvedilol 3.125 → 25 mg BID, or Metoprolol succinate 12.5–25 → 200 mg daily ◦ Spironolactone 12.5–25 mg daily (eGFR >30, K <5.0) ◦ Dapagliflozin or Empagliflozin 10 mg daily ◦ Furosemide 20–40 mg daily–BID, titrated to euvolemia • Monitor: BP, HR, SCr and K 1–2 weeks after each start/titration; weight and congestion each visit • Monitor MRA: K and SCr ~1 week, 1 month, 6 months; halve or stop if K >5.5

Question	Answer
Q5. Why add GDMT?	• ARNI/ACE inhibitor/ARB: lower mortality and hospitalization; reverse remodeling • Beta-blocker: mortality about 35% lower; fewer sudden deaths and hospitalizations • MRA: mortality 30% lower (RALES); fewer HF hospitalizations • SGLT2i: lower CV death or worsening HF (DAPA-HF, EMPEROR-Reduced) • Hydralazine/ISDN: lower mortality in African American patients, NYHA III–IV • Ivabradine, Vericiguat, Digoxin: reduce HF hospitalization, not mortality
Q6. What to check before starting Spironolactone?	• **eGFR >30 mL/min/1.73 m²** • **K <5.0 mEq/L** • SCr ≤2.5 mg/dL (men) or ≤2.0 mg/dL (women) • Other K-raising drugs: K supplements, NSAIDs, TMP-SMX
Q7. Side effects of Spironolactone?	• Hyperkalemia • Worsening renal function • Gynecomastia (~10%), breast tenderness, impotence, menstrual irregularity • Hyponatremia, volume depletion • Non-anion-gap metabolic acidosis
Q8. Other indications of Spironolactone?	• Primary hyperaldosteronism • Resistant hypertension • Ascites/edema in cirrhosis • Edema in nephrotic syndrome • Diuretic-induced hypokalemia • Hirsutism and acne (off-label)
Q9. Timing of monitoring for Spironolactone?	• K and SCr: about **1 week, 1 month and 6 months** after starting or a dose increase, then periodically • Restart the schedule when the ACE inhibitor/ARB dose increases • K >5.5: halve the dose or stop
Q10. Are you going to start full doses?	• No: start low and titrate • ACE inhibitor/ARB: double every 1–4 weeks • ARNI: double every 2–4 weeks • Beta-blocker: double every 1–2 weeks only when euvolemic • Aim: target or maximally tolerated dose
Q11. What are the targets?	• BP <130/80 mm Hg • GDMT at target or maximally tolerated doses • Euvolemia with fewer symptoms/better NYHA class • LVEF improvement (HFimpEF = EF >40% on follow-up)
Q12. What to modify in the patient's treatment?	• Stop Amlodipine: no HF benefit, causes peripheral edema that mimics congestion, and uses up BP room needed for GDMT • Control BP with GDMT (ARNI, beta-blocker once euvolemic, MRA, SGLT2i)
Also be ready for (exam add-ons)	• Goals: reduce mortality and HF hospitalization, relieve congestion and symptoms, prevent remodeling, improve exercise tolerance and quality of life • Non-pharm: sodium <2–3 g/day, fluid 1.5–2 L only if hyponatremia or advanced HF, daily weights, exercise or cardiac rehabilitation, vaccines (influenza, pneumococcal, COVID-19), stop smoking, limit alcohol, avoid NSAIDs • First-line: 4 pillars: ARNI Sacubitril/valsartan 24/26 mg BID (or ACE inhibitor/ARB), Bisoprolol 1.25 mg (or Carvedilol 3.125 mg BID, Metoprolol succinate 12.5–25 mg) when euvolemic, Spironolactone 12.5–25 mg (eGFR >30, K <5), Dapagliflozin 10 mg or Empagliflozin 10 mg; Furosemide 20–40 mg only for congestion • Pregnancy: ACE inhibitor, ARB, ARNI, MRA and SGLT2 inhibitors are contraindicated; use Metoprolol, Labetalol, Hydralazine + Isosorbide dinitrate and a loop diuretic; peripartum cardiomyopathy: add Bromocriptine and anticoagulation per specialist; counsel contraception • Renal/dialysis: check SCr and K (ARNI, ACE inhibitor, MRA); avoid MRA if eGFR <30 or K >5; Empagliflozin/Dapagliflozin can start at eGFR ≥20; higher loop-diuretic doses needed; Digoxin reduce in CKD • Liver: avoid ARNI in Child-Pugh C; Carvedilol avoid in severe disease; congestive hepatopathy improves with decongestion • GI (PUD/GERD): if on antiplatelet or anticoagulant add a PPI (Pantoprazole preferred with Clopidogrel); avoid NSAIDs • Counseling: ARNI: dizziness, angioedema, hyperkalemia, 36-hour ACE-inhibitor washout; β-blocker: fatigue, do not stop abruptly; Spironolactone: avoid potassium salt substitutes, gynecomastia; SGLT2 inhibitor: genital infection, hydration, hold on sick days; weigh daily and call if gain >1 kg in a day or 2 kg in a week

Case 4: HF + AF with QT Prolongation

Case details: HF and AF on Metoprolol, Digoxin, Dofetilide, Warfarin 5 mg daily, Simvastatin 40 mg daily; no HF therapy. INR 2.8. QT interval prolonged.

Question	Answer
Q1. Management of HF?	• Start GDMT: ARNI, MRA, SGLT2i • Beta-blocker: use Metoprolol succinate (not tartrate) titrated to target • Digoxin: keep level 0.5–0.9 ng/mL • Dofetilide with prolonged QT: check CrCl, K, Mg; reduce or stop if QTc >500 ms or rises >15% • Doses: ◦ Sacubitril/valsartan 24/26 mg BID (49/51 if on standard-dose ACE inhibitor/ARB) → 97/103 mg BID ◦ Metoprolol succinate 25 → 200 mg daily ◦ Spironolactone 12.5–25 mg daily (eGFR >30, K <5.0) ◦ Dapagliflozin or Empagliflozin 10 mg daily • Monitor: QTc and K/Mg (Dofetilide), Digoxin level, BP, SCr, K, weight • Non-pharm: sodium <2300 mg/day, daily weights, exercise/cardiac rehab, vaccines, avoid NSAIDs and non-DHP CCBs
Q2. Team wants to start azithromycin. Your action?	• Problem: additive **QT prolongation** with Dofetilide (torsades) • Preferred: choose a non-QT antibiotic for the infection (eg, Doxycycline for atypical coverage) • If no alternative: hold Dofetilide during the course with close QTc monitoring • Correct electrolytes: K >4, Mg >2 • Alternative dose: Doxycycline 100 mg PO BID • Monitor: ECG/QTc, K, Mg, infection response in 48–72 h
Q3. The team decided to add Amiodarone. Your action?	• Stop Dofetilide: two class III agents + prolonged QT • Warfarin: reduce dose by 30–50%; INR weekly • Digoxin: reduce dose by 30–50% (or reduce frequency); check level • Simvastatin: max 20 mg/day, or switch to Atorvastatin/Rosuvastatin • Metoprolol: watch for bradycardia and AV block
Q4. Monitoring parameters for Amiodarone?	• LFTs: baseline, then every 6 months • TFTs: baseline, then every 6 months • ECG: baseline, then periodically • Chest X-ray: baseline, then annually • PFTs with DLCO: baseline and if symptoms (stop if pulmonary fibrosis) • Eye exam: baseline if visual impairment, and if symptoms (stop if optic neuritis) • Skin: blue-gray discoloration, photosensitivity
Also be ready for (exam add-ons)	• Goals: reduce mortality and HF hospitalization, relieve congestion and symptoms, prevent remodeling, improve exercise tolerance and quality of life • Non-pharm: sodium <2–3 g/day, fluid 1.5–2 L only if hyponatremia or advanced HF, daily weights, exercise or cardiac rehabilitation, vaccines (influenza, pneumococcal, COVID-19), stop smoking, limit alcohol, avoid NSAIDs • First-line: 4 pillars: ARNI Sacubitril/valsartan 24/26 mg BID (or ACE inhibitor/ARB), Bisoprolol 1.25 mg (or Carvedilol 3.125 mg BID, Metoprolol succinate 12.5–25 mg) when euvolemic, Spironolactone 12.5–25 mg (eGFR >30, K <5), Dapagliflozin 10 mg or Empagliflozin 10 mg; Furosemide 20–40 mg only for congestion • Pregnancy: ACE inhibitor, ARB, ARNI, MRA and SGLT2 inhibitors are contraindicated; use Metoprolol, Labetalol, Hydralazine + Isosorbide dinitrate and a loop diuretic; peripartum cardiomyopathy: add Bromocriptine and anticoagulation per specialist; counsel contraception • Renal/dialysis: check SCr and K (ARNI, ACE inhibitor, MRA); avoid MRA if eGFR <30 or K >5; Empagliflozin/Dapagliflozin can start at eGFR ≥20; higher loop-diuretic doses needed; Digoxin reduce in CKD • Liver: avoid ARNI in Child-Pugh C; Carvedilol avoid in severe disease; congestive hepatopathy improves with decongestion • GI (PUD/GERD): if on antiplatelet or anticoagulant add a PPI (Pantoprazole preferred with Clopidogrel); avoid NSAIDs • Counseling: ARNI: dizziness, angioedema, hyperkalemia, 36-hour ACE-inhibitor washout; β-blocker: fatigue, do not stop abruptly; Spironolactone: avoid potassium salt substitutes, gynecomastia; SGLT2 inhibitor: genital infection, hydration, hold on sick days; weigh daily and call if gain >1 kg in a day or 2 kg in a week • Side scenario (QT): avoid QT-prolonging drugs (Azithromycin, Ondansetron, quinolones); keep K ≥4 and Mg ≥2; hold Dofetilide if QTc >500 ms; Amiodarone raises Digoxin and Warfarin levels (cut doses 30–50%) and Simvastatin max 20 mg

Case 5: Stage 2 HTN with Prediabetes ★ Repeated case, also in 2020

Case details: Stage 2 HTN (≥140/90) and prediabetes (A1C 5.7–6.4%, FBG 100–125, random BG 140–199). Smoker and obese.

Question	Answer
Q1. Patient risk factors for HTN?	• Obesity • Smoking • Prediabetes/insulin resistance • Sedentary lifestyle • High-sodium diet • Age, family history
Q2. Goal of therapy?	• BP ≤130/80 mm Hg; SBP ≤120 encouraged • Prevent complications (stroke, MI, HF, CKD) • Prevent progression of prediabetes to diabetes • Smoking cessation and weight loss
Q3. Non-pharmacological intervention?	• Weight loss ≥5%: about 1 mm Hg per 1 kg lost • DASH diet: 8–14 mm Hg • Sodium <2300 mg/day (ideally <1500): 6–8 mm Hg • Dietary K 3500–5000 mg/day: ~6 mm Hg • Alcohol: ideally none; ≤2 drinks/day men, ≤1 women (4–6 mm Hg) • Physical activity: structured aerobic and/or resistance exercise (4–10 mm Hg) • Stress reduction: 5–7 mm Hg • Smoking cessation • Drug therapy: stage 2 → start **two first-line agents from different classes** (ACE inhibitor/ARB, DHP CCB, thiazide), ideally a single pill ◦ Lisinopril 10 mg + Amlodipine 5 mg daily, or Losartan 50 mg + Chlorthalidone 12.5 mg daily • Monitor: BP in 4 weeks (home BP), SCr and K 1–2 weeks after ACE inhibitor/ARB or thiazide, Na, uric acid, glucose • Obesity and prediabetes: Semaglutide 2.4 mg SC weekly or Tirzepatide (BMI ≥30) gives weight loss, lowers BP and reduces progression to diabetes • If BP stays uncontrolled on three agents: add Spironolactone (resistant HTN); Aprocitentan (endothelin receptor antagonist) is a further option
Q4. Complications of DM and HTN?	• DM: DKA/HHS, hypoglycemia, diabetic foot infection, amputation, retinopathy, nephropathy, neuropathy, ASCVD • HTN: stroke, MI, HF, CKD, retinopathy, arrhythmia
Q5. Nicotine replacement: dosage forms? Patch and gum doses?	• Forms: patch, gum, lozenge, nasal spray, inhaler • Patch, >10 cigarettes/day: 21 mg/day × 6 weeks → 14 mg × 2 weeks → 7 mg × 2 weeks • Patch, ≤10 cigarettes/day: 14 mg/day × 6 weeks → 7 mg × 2 weeks • Gum strength: 4 mg if first cigarette within 30 min of waking; otherwise 2 mg • Gum schedule: 1 piece q1–2h weeks 1–6, q2–4h weeks 7–9, q4–6h weeks 10–12 • Gum use: at least 9 pieces/day, max 24/day; chew-and-park; no acidic drinks 15 min before • Best results: patch + short-acting form combined
Q6. Which vaccines are needed (no vaccine in the last 20 years)?	• Tdap once, then Td/Tdap every 10 years • Influenza yearly • Pneumococcal (smoker): one dose PCV20 or PCV21 (or PCV15 then PPSV23 ≥1 year later) • COVID-19 per current guidance • Hepatitis B if 19–59 years • By age: recombinant zoster ≥50 years; RSV per age and risk
Also be ready for (exam add-ons)	• Goals: BP <130/80, prevent stroke, MI, HF and kidney disease, control risk factors • Non-pharm: DASH diet (−8–14 mm Hg), sodium <1.5–2.3 g/day (−2–8), weight loss (−1 mm Hg per kg), exercise 150 min/week (−4–9), alcohol limit, smoking cessation, stress management • First-line: stage 2: two agents (ACE inhibitor/ARB + CCB or thiazide) e.g., Lisinopril 10 mg + Amlodipine 5 mg or Hydrochlorothiazide 12.5–25 mg; DM or albuminuria: ACE inhibitor/ARB • Pregnancy: avoid ACE inhibitors, ARBs, direct renin inhibitors and thiazides in planning; use Labetalol 100–400 mg BID, Nifedipine ER 30–60 mg or Methyldopa; low-dose Aspirin from 12 weeks if preeclampsia risk • Renal/dialysis: ACE inhibitor/ARB for albuminuria (monitor SCr up to 30% rise and K); thiazides weaker at eGFR <30 (use loop); avoid K-sparing in advanced CKD • Liver: Labetalol hepatotoxicity monitoring; avoid non-selective β-blockers in decompensated cirrhosis (except Carvedilol for varices) • HF/DM: ARNI, β-blocker, MRA and SGLT2 inhibitor in HFrEF; avoid non-DHP CCB in HFrEF and Pioglitazone • Counseling: home BP monitoring, adherence, ACE inhibitor cough/angioedema, Amlodipine edema, thiazide gout/hypokalemia, rise slowly, avoid NSAIDs/decongestants/licorice • Smoking cessation: counsel (5 A's); Varenicline 0.5 mg daily × 3 days, 0.5 mg BID × 4 days, then 1 mg BID × 12 weeks, or combination NRT (patch 21/14/7 mg + gum/lozenge 2–4 mg); Bupropion SR 150 mg daily × 3 days then 150 mg BID, also first-line (Varenicline is the most effective single agent); monitor BP on NRT and Bupropion

Question	Answer
	• Vaccines: influenza yearly, pneumococcal (PCV20/21), Td/Tdap every 10 years, COVID-19, hepatitis B, zoster

Case 6: Dyslipidemia ★ Repeated case, also in 2018–2019

Case details: Patient <75 years with dyslipidemia; 10-year ASCVD risk 14%.

Question	Answer
Q1. What are the goals of therapy?	• Reduce ASCVD events (MI, stroke, PAD) and mortality • LDL-C reduction 30–49% with moderate-intensity statin • If TG ≥500 mg/dL: prevent pancreatitis
Q2. Non-pharmacological therapy?	• Heart-healthy diet: DASH or Mediterranean; fruits, vegetables, whole grains, low-fat dairy • Limit: sweets, sugary drinks, red meat; replace saturated with unsaturated fat • Exercise: moderate–vigorous aerobic 3–4 times/week, ~40 min/session • Weight loss and smoking cessation
Q3. Which statin to start, and dose?	• Risk 14% = intermediate (7.5–19.9%): moderate-intensity statin if risk discussion favors (risk enhancers; CAC if uncertain) • Options: ◦ Atorvastatin 10–20 mg ◦ Rosuvastatin 5–10 mg ◦ Simvastatin 20–40 mg ◦ Pravastatin 40–80 mg ◦ Lovastatin 40–80 mg ◦ Fluvastatin XL 80 mg ◦ Pitavastatin 1–4 mg • Monitor: lipid panel 4–12 weeks, LFT baseline, muscle symptoms
Q4. Benefits of statins?	• LDL-C: ↓30–49% (moderate), ↓≥50% (high intensity) • TG: modest ↓; HDL: small ↑ • Outcomes: fewer MI, stroke, revascularization and CV deaths
Q5. PCSK9 inhibitors: names and uses? Should they be added?	• Evolocumab: 140 mg SC q2 weeks or 420 mg monthly • Alirocumab: 75–150 mg SC q2 weeks or 300 mg q4 weeks • Inclisiran (siRNA): 284 mg SC at 0 and 3 months, then every 6 months • Uses: clinical ASCVD above goal on max statin ± Ezetimibe; heterozygous FH; LDL-C ≥190 • This patient: No (primary prevention, intermediate risk); statin first, then Ezetimibe if needed • **Newer oral option:** Bempedoic acid 180 mg daily (ATP citrate lyase inhibitor) for statin intolerance or as add-on to Ezetimibe; lowers MACE (CLEAR Outcomes); limit Simvastatin to 20 mg and Pravastatin to 40 mg; may raise uric acid, tendon-rupture warning • Triglycerides: Icosapent ethyl 2 g BID if TG 135–499 mg/dL with ASCVD, or DM plus risk factors, on a statin (REDUCE-IT); Olezarsen (APOC3 antisense) for familial chylomicronemia syndrome
Q6. Monitoring plan for statins?	• Efficacy: lipid panel 4–12 weeks after start or change, then every 3–12 months • LFTs: baseline, then only if clinically indicated • CK: only if muscle symptoms • Mild–moderate muscle symptoms: hold, re-evaluate, rechallenge lower dose or different statin • Stop: **CK >10× ULN** with symptoms or rhabdomyolysis; persistent ALT >3× ULN
Q7. Blood pressure target?	• <130/80 mm Hg
Q8. Is Aspirin indicated?	• No: not for routine primary prevention (bleeding offsets benefit) • Yes for secondary prevention (established ASCVD)
Also be ready for (exam add-ons)	• Goals: lower LDL-C and ASCVD risk (primary: ≥30–49% reduction by risk; secondary: ≥50% and <70 mg/dL), reduce events and mortality; prevent pancreatitis if TG very high • Non-pharm: Mediterranean diet (saturated fat <7%, no trans fat), fiber 10–25 g, plant sterols, exercise 150 min/week, weight loss, smoking cessation, limit alcohol and sugar • First-line: risk 7.5–<20%: moderate-intensity statin (Atorvastatin 10–20 mg, Rosuvastatin 5–10 mg); ≥20% or ASCVD: high-intensity (Atorvastatin 40–80 mg, Rosuvastatin 20–40 mg); add Ezetimibe 10 mg then a PCSK9 inhibitor (Evolocumab 140 mg q2 weeks, Alirocumab 75–150 mg q2 weeks) if LDL above goal; TG >500: Fenofibrate • Pregnancy: statins and PCSK9 inhibitors are stopped before conception or as soon as pregnancy is confirmed (and during breastfeeding); Ezetimibe is avoided (limited data); Colesevelam is acceptable • Renal/dialysis: Atorvastatin and Fluvastatin need no adjustment; Rosuvastatin max 10 mg if CrCl <30; Simvastatin/Pravastatin start low; avoid Gemfibrozil with statins; Fenofibrate reduce • Liver: avoid in active liver disease; ALT >3× ULN: hold; do not use in decompensated cirrhosis • Interactions: Amiodarone → Simvastatin ≤20 mg; Diltiazem/Verapamil → Simvastatin ≤10 mg; Clarithromycin, azoles, Cyclosporine; Gemfibrozil + statin = rhabdomyolysis • Counseling: evening dosing for short half-life statins (Simvastatin, Lovastatin, Pravastatin), muscle pain with dark urine = stop and call, avoid large amounts of grapefruit juice (Simvastatin, Atorvastatin), adherence, diet, lipid panel 4–12 weeks then every 3–12 months

Case 7: ACS (STEMI, cath lab available) ★ Repeated case, also in 2025, 2022–2023, 2018–2019

Case details: Patient presented to the ER; ECG shows STEMI. Facility has a cath lab.

Question	Answer
Q1. What will you give in the ER?	• Aspirin: 162–325 mg chewed, non-enteric coated • P2Y12 load: Ticagrelor 180 mg or Prasugrel 60 mg (preferred), or Clopidogrel 600 mg • Nitroglycerin: 0.4 mg SL q5min × 3; IV 10 mcg/min if persistent ischemia, HF or HTN • Oxygen if SaO2 <90% • Morphine: 2–4 mg IV if pain persists despite anti-ischemic therapy • Oral beta-blocker within 24 h if no contraindication • High-intensity statin early • Monitor: continuous ECG, BP, HR, SaO2, pain, troponin, Hgb, SCr
Q2. What will you give inside the cath lab?	• UFH: 70–100 units/kg IV bolus without GP IIb/IIIa, or 50–70 units/kg with GP IIb/IIIa, titrated to ACT • Bivalirudin: 0.75 mg/kg IV, then 1.75 mg/kg/h (preferred if HIT) • GP IIb/IIIa (bailout or high thrombus burden): ◦ Eptifibatide 180 mcg/kg IV × 2 (10 min apart), then 2 mcg/kg/min ◦ Tirofiban 25 mcg/kg IV, then 0.15 mcg/kg/min • Cangrelor 30 mcg/kg then 4 mcg/kg/min if P2Y12-naive • Monitor: ACT during PCI, bleeding, platelets (GP IIb/IIIa), SCr
Q3. How to reduce the risk of contrast-induced AKI?	• Volume expansion: IV isotonic saline (bicarbonate adds no benefit, PRESERVE); start 6–12 h before when time allows • Target urine output >150 mL/h • Hold Metformin ≥48 h after contrast • Stop nephrotoxins and avoid diuretics • Contrast: low- or iso-osmolar, smallest volume • Acetylcysteine: optional oral add-on (weak); no benefit over hydration • Do not use: theophylline, prophylactic dialysis • Monitor: SCr at 24–72 h after contrast, urine output, K, volume status (avoid overload in HF)
Q4. What will you give upon discharge?	• DAPT ≥12 months: Aspirin 81 mg + Ticagrelor 90 mg BID (or Prasugrel 10 mg, Clopidogrel 75 mg); then Aspirin indefinitely • Atorvastatin 80 mg ± Ezetimibe • Beta-blocker: Metoprolol succinate, Carvedilol or Bisoprolol • ACE inhibitor: Lisinopril 2.5–5 mg daily if EF ≤40%, HTN, DM or CKD • MRA: Eplerenone 25 → 50 mg daily if EF ≤40% with HF or DM • SL Nitroglycerin PRN, PPI if bleeding risk, cardiac rehabilitation • Monitor: chest pain, bleeding (Hgb), BP, HR, SCr/K (ACE inhibitor), lipids 4–8 weeks, LFT baseline, adherence to DAPT • Non-pharm: smoking cessation, heart-healthy diet, weight, exercise, cardiac rehab
Also be ready for (exam add-ons)	• Goals: restore infarct-artery patency quickly (door-to-balloon ≤90 min, door-to-needle ≤30 min), limit infarct size, relieve ischemia, prevent arrhythmia, heart failure and reinfarction • Non-pharm: primary PCI (preferred), cardiac rehab, smoking cessation, Mediterranean diet, exercise 150 min/week, weight control, sodium restriction, vaccines (influenza), driving advice • First-line: Aspirin 162–325 mg chewed, P2Y12 inhibitor (Ticagrelor 180 mg or Prasugrel 60 mg or Clopidogrel 600 mg), UFH 70–100 units/kg in the cath lab, Atorvastatin 80 mg, oral β-blocker within 24 h if LVEF reduced, HF or ventricular arrhythmia and no shock (2025 ACC/AHA; reassess after 1 year if LVEF is preserved), Nitroglycerin SL for pain • Pregnancy: MI in pregnancy is rare: PCI preferred, avoid fibrinolysis if possible; low-dose Aspirin and Clopidogrel acceptable; avoid ACE inhibitors/ARBs/MRA and statins; Metoprolol and Labetalol are safe; UFH or Enoxaparin acceptable • Renal/dialysis: contrast nephropathy prevention (hydration, minimal contrast, hold Metformin and nephrotoxins); Enoxaparin 1 mg/kg q24h if CrCl <30; Bivalirudin: reduce the infusion if CrCl <30; Ticagrelor needs no adjustment • Liver: avoid Ticagrelor in severe hepatic impairment; statin caution when ALT >3× ULN; bleeding risk higher • HF/shock: avoid β-blocker in acute HF or shock; add Spironolactone if EF <40% with DM or HF; ACE inhibitor/ARNI for EF <40% • GI (PUD/GERD): add PPI with DAPT in patients at GI-bleeding risk (Pantoprazole; avoid Omeprazole with Clopidogrel) • LDL-C goal <55 mg/dL (add Ezetimibe if ≥70 on maximal statin, 2025 ACC/AHA); Counseling: DAPT adherence for 12 months (never stop early), bleeding signs, Ticagrelor dyspnea and Aspirin dose ≤100 mg, Nitroglycerin use (call 911 if pain persists 5 min after first dose), statin muscle pain, cardiac rehab, return to activity

Case 8: NSTEMI with ADHF and Hypertensive Emergency ★ Repeated case, also in 2015

Case details: 65-year-old woman with chest pain. PMH: HTN, DM, DLP, CAD with DES 2007. Troponin normal, ↑BUN, ↑SCr, ↑A1C, ECHO: reduced EF (HFrEF). ECG:

ST depression. CXR: pulmonary edema. BP 160–210 systolic. On Diltiazem for AF.

Question	Answer
Q1. What is the problem list?	• NSTE-ACS: chest pain + ST depression (repeat troponin) • ADHF (warm and wet): reduced EF + pulmonary edema • Hypertensive emergency: severe BP with target-organ damage (ACS, pulmonary edema, kidney) • AKI/CKD: ↑BUN, ↑SCr • Uncontrolled DM: ↑A1C • AF on a non-DHP CCB • Dyslipidemia with established ASCVD
Q2. How to manage her ischemia and HTN?	• Agent: IV Nitroglycerin 10 mcg/min, titrate (max 200 mcg/min); treats ischemia, preload and BP • Avoid Nitroprusside: coronary steal in ACS; thiocyanate accumulates with renal impairment • BP lowering: reduce SBP by **no more than 25% in the first hour**, then to 160/100–110 over 2–6 h, then to normal over 24–48 h • ACS therapy: Aspirin, P2Y12 inhibitor, anticoagulant (UFH preferred with renal impairment), statin • Monitor: BP every 5–15 min while titrating, headache, HR, chest pain, ECG, troponin
Q3. How to manage her pulmonary edema? Monitoring?	• IV loop: Furosemide 40 mg IV (or 1–2.5× home oral dose IV), bolus BID or infusion • Double dose until response (eg UOP >500 mL in 2 h); add Metolazone if refractory • Adjuncts: IV Nitroglycerin, oxygen/non-invasive ventilation • Monitor: UOP, intake/output, daily weight, SCr/BUN, K, Mg, Na, BP, oxygenation, symptoms
Q4. What is the target LDL?	• Established ASCVD: LDL-C <70 mg/dL; <55 mg/dL is reasonable (very high risk)
Q5. At which ALT and CK levels will you hold or discontinue statins?	• ALT: persistent >3× ULN → stop and evaluate • CK: **>10× ULN** with symptoms or rhabdomyolysis → stop • Mild–moderate muscle symptoms: hold, then rechallenge lower dose or different statin
Q6. On rate control (Diltiazem) with AF: how to control her AF? Dose? Monitoring?	• Stop Diltiazem: non-DHP CCB is harmful in HFrEF • No beta-blocker now: she is decompensated; start once euvolemic • Amiodarone: IV 150 mg over 10 min, 1 mg/min × 6 h, then 0.5 mg/min × 18 h; oral load 400 mg BID–TID to ~6–10 g total, then 200 mg daily • Digoxin IV is an alternative for rate control • Target HR: <80 at rest (HF) • Anticoagulation: CHA2DS2-VASc high (CHF, HTN, DM, vascular, female, age 65) • Monitor Amiodarone: LFTs and TFTs every 6 months, ECG, CXR yearly, PFTs and eye exam as indicated
Q7. Maximum dose of Simvastatin with Amiodarone?	• Simvastatin 20 mg/day • Better: switch to Atorvastatin 80 mg (ASCVD needs high intensity)
Q8. Duration of DAPT with DES, medical management, and BMS?	• ACS (any stent type or medically managed): DAPT **≥12 months** • Then: Aspirin indefinitely • Stable CAD with DES (not ACS): 6 months
Q9. Which antiarrhythmics to avoid, and why?	• Class Ic (Flecainide, Propafenone): proarrhythmia, higher mortality in structural heart disease • Dronedarone: doubled mortality in NYHA III–IV or recent decompensation • Class Ia (Disopyramide): negative inotrope • Sotalol: caution; avoid with severely reduced EF • Safe in HF: Amiodarone and Dofetilide
Also be ready for (exam add-ons)	• Goals: relieve ischemia, prevent MI and death, treat HF and hypertension, secondary prevention • Non-pharm: early invasive strategy by risk (GRACE), cardiac rehab, smoking cessation, diet (Mediterranean, sodium restriction), exercise, weight control, vaccines • First-line: Aspirin 162–325 mg, P2Y12 inhibitor (Ticagrelor 180 mg), Enoxaparin 1 mg/kg q12h or UFH, oral β-blocker early if reduced LVEF or HF without shock (reassess at 1 year if LVEF preserved), Nitroglycerin 5–10 mcg/min for persistent ischemia, hypertension or pulmonary edema; Atorvastatin 80 mg • Pregnancy: ACE inhibitor/ARB/MRA/statin contraindicated; Metoprolol, Labetalol and Nifedipine are used; low-dose Aspirin and Clopidogrel are acceptable; amiodarone avoided • Renal/dialysis: avoid Sodium nitroprusside (cyanide/thiocyanate); Enoxaparin 1 mg/kg q24h if CrCl <30; contrast prevention; loop diuretic higher doses • Liver: statin and Amiodarone hepatotoxicity monitoring; Carvedilol avoid in severe liver disease • HF (ADHF): Furosemide 20–40 mg IV (or 2.5× the home dose); avoid non-DHP CCB and Flecainide/Propafenone/Sotalol/Dronedarone; Amiodarone or Dofetilide for AF • GI (PUD/GERD): PPI with DAPT or anticoagulant; avoid NSAIDs • Counseling: DAPT 12 months, statin adherence (stop and call for severe muscle pain), Nitroglycerin use, daily weights, BP log, warning signs of MI/HF

Case 9: STEMI

Case details: 68-year-old man with chest pain radiating to the left arm. PMH: smoker, HTN, DM. Home: Lisinopril 20 mg/day, Fluvastatin 20 mg/day, Metformin 1 g BID, Amlodipine 10 mg/day, Insulin Mixtard 40 units SC TID. ECG: ST elevation. Febrile, tachypneic, tachycardic. Weight 86 kg, height 175 cm. Troponin normal, ↑BG,

↑A1C, SCr 109 µmol/L.

Question	Answer
Q1. What is the diagnosis?	• STEMI: ST elevation + typical radiating chest pain • Early troponin may be normal; do not delay reperfusion waiting for it
Q2. Risk factors that contributed?	• Smoking • HTN • Uncontrolled DM • Age and male sex • Dyslipidemia (on low-intensity statin) • Overweight (BMI ~28)
Q3. Complications of STEMI?	• Cardiogenic shock • HF/LV dysfunction • Arrhythmias (VT/VF, AF, heart block) • Mechanical: papillary muscle rupture (acute MR), VSD, free-wall rupture/tamponade • LV thrombus → embolic stroke • Pericarditis • LV aneurysm • Reinfarction
Q4. Management if transferred to a hospital with a cath lab?	• Primary PCI: first medical contact to device ≤120 min when transferring (≤90 min at a PCI center) • Aspirin: 162–325 mg chewed • P2Y12: Ticagrelor 180 mg or Prasugrel 60 mg (age <75, weight ≥60 kg, no stroke/TIA), or Clopidogrel **600 mg** (300 mg is the dose with fibrinolysis) • Anticoagulant: UFH 70–100 units/kg IV in the lab; Bivalirudin 0.75 mg/kg then 1.75 mg/kg/h if HIT • Analgesia: Nitroglycerin 0.4 mg SL q5min × 3, IV if persistent; Morphine 2–4 mg IV only if refractory • Oxygen if SaO2 <90% • Hold Metformin around contrast; manage glucose with insulin (140–180 mg/dL) • Monitor: ECG, BP, HR, SaO2, pain, glucose every 1–2 h on insulin, SCr, bleeding
Q5. Management after PCI?	• Aspirin 81 mg daily indefinitely • Ticagrelor 90 mg BID (or Prasugrel 10 mg) for ≥12 months • Beta-blocker: Metoprolol succinate or Bisoprolol; reassess after 1 year if EF >50% and no other indication • Switch Fluvastatin to Atorvastatin 80 mg • Continue Lisinopril • Monitor: chest pain, bleeding (Hgb), BP, HR, SCr/K (ACE inhibitor), lipids 4–8 weeks, LFT baseline, adherence to DAPT
Q6. Long-term treatment upon discharge?	• DAPT, high-intensity statin ± Ezetimibe, beta-blocker, ACE inhibitor, SL Nitroglycerin PRN • Amlodipine: keep only if BP or angina still needs it after beta-blocker + ACE inhibitor • Diabetes: add an agent with CV benefit and intensify insulin plan ◦ Semaglutide 0.25 mg SC weekly × 4 weeks → 0.5 → 1 mg ◦ or Liraglutide 0.6 → 1.2 → 1.8 mg daily, or Dulaglutide 0.75 → 1.5 mg weekly ◦ or Empagliflozin 10 mg daily ◦ consider converting Mixtard to basal-bolus insulin • Resume Metformin 48 h after contrast if renal function stable • Smoking cessation, cardiac rehabilitation • Doses: ◦ Aspirin 81 mg daily (indefinitely) ◦ Ticagrelor 90 mg BID or Clopidogrel 75 mg daily for 12 months ◦ Atorvastatin 80 mg daily ± Ezetimibe 10 mg ◦ Metoprolol succinate 25 → 200 mg daily or Bisoprolol 1.25 → 10 mg daily ◦ Lisinopril 2.5–5 → 20–40 mg daily ◦ Nitroglycerin 0.4 mg SL q5min × 3 PRN chest pain • Monitor: chest pain, bleeding (Hgb), BP, HR, SCr/K (ACE inhibitor), lipids 4–8 weeks, LFT baseline, adherence to DAPT • Monitor DM: A1C every 3 months, SMBG, GI effects of GLP-1 RA, SCr before Metformin restart
Also be ready for (exam add-ons)	• Goals: restore infarct-artery patency quickly (door-to-balloon ≤90 min, door-to-needle ≤30 min), limit infarct size, relieve ischemia, prevent arrhythmia, heart failure and reinfarction • Non-pharm: primary PCI (preferred), cardiac rehab, smoking cessation, Mediterranean diet, exercise 150 min/week, weight control, sodium restriction, vaccines (influenza), driving advice • First-line: Aspirin 162–325 mg chewed, P2Y12 inhibitor (Ticagrelor 180 mg or Prasugrel 60 mg or Clopidogrel 600 mg), UFH 70–100 units/kg in the cath lab, Atorvastatin 80 mg, oral β-blocker within 24 h if LVEF reduced, HF or ventricular arrhythmia and no shock (2025 ACC/AHA; reassess after 1 year if LVEF is preserved), Nitroglycerin SL for pain • Pregnancy: MI in pregnancy is rare: PCI preferred, avoid fibrinolysis if possible; low-dose Aspirin and Clopidogrel acceptable; avoid ACE inhibitors/ARBs/MRA and statins; Metoprolol and Labetalol are safe; UFH or Enoxaparin acceptable • Renal/dialysis: contrast nephropathy prevention (hydration, minimal contrast, hold Metformin and nephrotoxins); Enoxaparin 1 mg/kg q24h if CrCl <30; Bivalirudin: reduce the infusion if CrCl <30;

Question	Answer
	Ticagrelor needs no adjustment • Liver: avoid Ticagrelor in severe hepatic impairment; statin caution when ALT >3× ULN; bleeding risk higher • HF/shock: avoid β-blocker in acute HF or shock; add Spironolactone if EF <40% with DM or HF; ACE inhibitor/ARNI for EF <40% • GI (PUD/GERD): add PPI with DAPT in patients at GI-bleeding risk (Pantoprazole; avoid Omeprazole with Clopidogrel) • LDL-C goal <55 mg/dL (add Ezetimibe if ≥70 on maximal statin, 2025 ACC/AHA); Counseling: DAPT adherence for 12 months (never stop early), bleeding signs, Ticagrelor dyspnea and Aspirin dose ≤100 mg, Nitroglycerin use (call 911 if pain persists 5 min after first dose), statin muscle pain, cardiac rehab, return to activity

Case 10: VTE in a Cancer Patient

Case details: Cancer patient, post-surgery, obese, with a long flight.

Question	Answer
Q1. Risk factors for VTE?	• Active cancer (± chemotherapy) • Recent surgery • Obesity • Immobility (long flight >8 h)
Q2. Pharmacological and non-pharmacological prophylaxis?	• Enoxaparin: 40 mg SC daily; BMI ≥40: 40 mg SC BID • Alternative: UFH 5000 units SC q8–12h • After major abdominal/pelvic cancer surgery: extend LMWH to 4 weeks • Mechanical (if bleeding risk): intermittent pneumatic compression ≥18 h/day; early ambulation; graduated stockings have limited efficacy • Monitor: platelets (HIT), SCr (CrCl), bleeding, anti-Xa only in obesity/renal impairment
Q3. Acute PE management?	• Hemodynamically stable: Enoxaparin 1 mg/kg SC q12h, or Apixaban 10 mg BID × 7 days • Massive (unstable): UFH 80 units/kg IV then 18 units/kg/h (short half-life, easy to stop for lysis) + Alteplase 100 mg over 2 h • Obesity: anti-Xa peak 4 h after the 3rd–4th dose; target 0.6–1.0 units/mL (q12h dosing) • Monitor: aPTT/anti-Xa every 6 h until stable (UFH), Hgb, platelets, BP/HR, SaO2, RV function
Q4. Patient is for discharge: plan for PE?	• Preferred (CHEST 2021): oral factor Xa inhibitor: Apixaban, Edoxaban or Rivaroxaban • Prefer LMWH or Apixaban with luminal GI cancer (bleeding with Edoxaban/Rivaroxaban) • LMWH also preferred with GU cancer, active mucosal lesions or interacting chemotherapy • Doses: ◦ Apixaban 10 mg BID × 7 days, then 5 mg BID ◦ Rivaroxaban 15 mg BID × 21 days, then 20 mg daily ◦ Edoxaban 60 mg daily after ≥5 days of parenteral (30 mg if CrCl 15–50 or ≤60 kg) ◦ Enoxaparin 1 mg/kg SC q12h (or Dalteparin 200 units/kg daily × 1 month, then 150 units/kg) • Monitor: CBC, SCr, LFTs, bleeding, drug interactions with chemotherapy, recurrence symptoms
Q5. What is the duration?	• At least 3–6 months • Continue indefinitely while cancer is active or under treatment
Q6. Cancer cured, second VTE: your plan?	• Apixaban: 10 mg BID × 7 days, then 5 mg BID; after 6 months may reduce to 2.5 mg BID • Rivaroxaban: 15 mg BID × 21 days, then 20 mg daily; after 6 months may reduce to 10 mg daily
Q7. Duration for the second VTE?	• Indefinite, with yearly reassessment of bleeding risk
Also be ready for (exam add-ons)	• Goals: prevent VTE in high-risk patients, treat established VTE, prevent recurrence and bleeding • Non-pharm: early mobilization, intermittent pneumatic compression ≥18 h/day, elastic stockings (limited efficacy), hydration and calf exercises on long flights, weight loss • First-line: prophylaxis Enoxaparin 40 mg daily (BMI ≥40: 40 mg BID); treatment Enoxaparin 1 mg/kg q12h (actual body weight, no routine dose cap in obesity) or Apixaban 10 mg BID × 7 days then 5 mg BID in cancer (Rivaroxaban, Edoxaban acceptable); ≥6 months • Pregnancy: LMWH only; Warfarin and DOACs contraindicated; continue through 6 weeks postpartum; stop Enoxaparin 24 h before neuraxial anesthesia (prophylactic dose 12 h) • Renal/dialysis: Enoxaparin 30 mg daily (prophylaxis) or 1 mg/kg daily if CrCl <30; UFH in dialysis; Apixaban dose unchanged for VTE; anti-Xa 0.5–1 IU/mL for treatment • Liver: avoid DOACs in Child-Pugh C; monitor platelets • Bleeding risk (cancer, GI/GU tumors): Dabigatran, Edoxaban and Rivaroxaban raise GI bleeding; prefer Apixaban; hold anticoagulation if platelets <50,000 (reduce dose 25–50,000) • Counseling: injection technique, bleeding signs, avoid NSAIDs, medical alert, perioperative holds, duration and re-evaluation, hydration on flights

Case 11: HTN with DM and Dyslipidemia

Case details: BP 160/90, A1C 9.5%, high TG and LDL-C, 10-year ASCVD risk 5.4%. Home: Hydrochlorothiazide 25 mg.

Question	Answer
Q1. What is the problem list?	• Stage 2 HTN, uncontrolled • Type 2 DM, uncontrolled • Mixed dyslipidemia
Q2. Management for HTN?	• Add ACE inhibitor or ARB (DM; especially with albuminuria): Lisinopril 10 mg daily • Continue Hydrochlorothiazide 25 mg; consider a single-pill combination • Goal: <130/80 • Follow-up: K and SCr 1–2 weeks; BP within 4 weeks
Q3. Non-pharmacological management for HTN?	• Weight loss ≥5% • DASH diet • Sodium <2300 mg/day (ideally <1500) • Dietary K 3500–5000 mg/day • Limit alcohol • Regular aerobic and resistance exercise • Stress reduction • Smoking cessation
Q4. Management of HHS (glucose >600, pH >7.3, HCO3 >15, no ketones)?	• Fluids: 0.9% NaCl 15–20 mL/kg (1–1.5 L) first hour, then 250–500 mL/h; 0.45% NaCl if Na normal/high • Insulin: regular insulin IV 0.1 unit/kg/h (± 0.1 unit/kg bolus); only if K ≥3.3 • At BG 300: reduce to 0.02–0.05 unit/kg/h, add dextrose, keep BG 200–300 until mental status and osmolality resolve • Potassium: K <3.3 hold insulin and give K; 3.3–5.2 add 20–30 mEq/L; >5.2 none, monitor • Transition (insulin-naive): SC 0.5–0.8 units/kg/day; overlap basal 1–2 h before stopping the drip ◦ 70 kg × 0.5 = 35 units/day → Insulin glargine ~17 units daily + Insulin aspart 6 units TID before meals
Q5. Management for dyslipidemia?	• DM aged 40–75: moderate-intensity statin regardless of risk score ◦ Atorvastatin 10–20 mg or Rosuvastatin 5–10 mg • High intensity if multiple ASCVD risk factors or DM-specific enhancers • TG: improve glycemic control first; if ≥500 mg/dL add fibrate or omega-3 to prevent pancreatitis • Monitor: lipid panel 4–12 weeks, then every 3–12 months; LFT baseline; muscle symptoms
Also be ready for (exam add-ons)	• Goals: BP <130/80, prevent stroke, MI, HF and kidney disease, control risk factors • Non-pharm: DASH diet (−8–14 mm Hg), sodium <1.5–2.3 g/day (−2–8), weight loss (−1 mm Hg per kg), exercise 150 min/week (−4–9), alcohol limit, smoking cessation, stress management • First-line: stage 2: two agents (ACE inhibitor/ARB + CCB or thiazide) e.g., Lisinopril 10 mg + Amlodipine 5 mg or Hydrochlorothiazide 12.5–25 mg; DM or albuminuria: ACE inhibitor/ARB • Pregnancy: avoid ACE inhibitors, ARBs, direct renin inhibitors and thiazides in planning; use Labetalol 100–400 mg BID, Nifedipine ER 30–60 mg or Methyldopa; low-dose Aspirin from 12 weeks if preeclampsia risk • Renal/dialysis: ACE inhibitor/ARB for albuminuria (monitor SCr up to 30% rise and K); thiazides weaker at eGFR <30 (use loop); avoid K-sparing in advanced CKD • Liver: Labetalol hepatotoxicity monitoring; avoid non-selective β-blockers in decompensated cirrhosis (except Carvedilol for varices) • HF/DM: ARNI, β-blocker, MRA and SGLT2 inhibitor in HFrEF; avoid non-DHP CCB in HFrEF and Pioglitazone • Counseling: home BP monitoring, adherence, ACE inhibitor cough/angioedema, Amlodipine edema, thiazide gout/hypokalemia, rise slowly, avoid NSAIDs/decongestants/licorice • DM: Metformin 500 mg daily titrated to 2 g; add GLP-1 RA or SGLT2 inhibitor if ASCVD/CKD/HF; HHS: IV fluids and Insulin infusion; basal-bolus Insulin at discharge (0.5 units/kg TDD: half basal) • Dyslipidemia: moderate-intensity statin (age 40–75 with DM) or high-intensity with risk factors; LDL <70 (<55 with ASCVD)

Case 12: Anticoagulation (Warfarin with Nasal Bleeding) ★ Repeated case, also in 2018–2019

Case details: Patient on Warfarin presented to the clinic with nasal bleeding.

Question	Answer
Q1. What is your action?	• Assess severity: vitals, Hgb, INR; minor vs major bleeding • Minor epistaxis: local pressure/topical measures; hold Warfarin ◦ INR 4.5–10: may give oral Vitamin K 2.5–5 mg; check INR daily ◦ INR >10: oral Vitamin K 2.5–5 mg; recheck in 12–24 h • Major/life-threatening: **Vitamin K 5–10 mg slow IV + 4F-PCC** • Find the cause: new drugs, diet, illness, dosing error, adherence • Resume at a 5–20% lower weekly dose once INR is in range; recheck INR in 3–7 days • If the patient were on a DOAC: Idarucizumab 5 g IV for Dabigatran (humanized antibody fragment); 4F-PCC (or Andexanet alfa where available) for Apixaban and Rivaroxaban

Question	Answer
Also be ready for (exam add-ons)	• Goals: keep INR in range (2–3 for most indications; 2.5–3.5 mechanical mitral valve), stop bleeding, avoid thrombosis • Non-pharm: local pressure (10–15 min) for nasal bleeding, identify the cause (dose, interacting drug, infection, diet, alcohol), consistent Vitamin K intake, written dose calendar • First-line: INR >4.5–10 without bleeding: hold 1–2 doses, reduce weekly dose 10–20%; no routine Vitamin K unless high bleeding risk (Vitamin K 1–2.5 mg PO); major bleeding: Vitamin K 5–10 mg IV + 4-factor PCC • Pregnancy: Warfarin is teratogenic (6–12 weeks) and crosses the placenta: switch to LMWH before week 6 (mechanical valves: specialist decision); Warfarin is safe during breastfeeding • Renal/dialysis: higher bleeding risk, lower dose requirement; consider Apixaban in AF with dialysis (non-valvular) • Liver: baseline INR high, lower doses, closer monitoring • HF: congestion raises INR sensitivity; interacting drugs: Amiodarone (30–50% dose cut), Fluconazole, Metronidazole, TMP-SMX, quinolones, Rifampin (increase dose), Carbamazepine • Counseling: bleeding signs, steady greens, avoid alcohol binges and NSAIDs/Aspirin, report new drugs/antibiotics, never double a missed dose, INR 3–5 days after any change, carry a card

Case 13: ADHF with AF, HFrEF (EF 20%) and TPN

Case details: Congested and warm patient with ADHF and AF, HFrEF (EF 20%), GI surgery with NPO for 7 days, and depression. Weight 80 kg (used in the TPN calculation).

Question	Answer
Q1. What is the diagnosis?	• **Acute decompensated HFrEF** with AF; hemodynamic profile **warm and wet** (Stevenson profile B, "subset 2")
Q2. What is the treatment of subset 2 (warm and wet)?	• **IV loop diuretic:** Furosemide 40–80 mg IV (or 2–2.5× the home oral dose); check urine output >100–150 mL/h and spot urine sodium >50–70 mEq/L at 2 h, then double the dose if inadequate • **IV vasodilator** if SBP >110: Nitroglycerin 10–20 mcg/min, titrate to 200 mcg/min; oxygen if SpO2 <90%, NIV for pulmonary edema • Continue GDMT unless shock or hyperkalemia
Q3. Which medicines have a mortality benefit?	• ARNI (or ACE inhibitor/ARB), evidence-based β-blocker (Bisoprolol, Carvedilol, Metoprolol succinate), MRA (Spironolactone, Eplerenone), SGLT2 inhibitor (Dapagliflozin, Empagliflozin), Hydralazine + Isosorbide dinitrate (African American patients or ACEi/ARNI intolerance) • Vericiguat and Ivabradine reduce HF hospitalization; Digoxin reduces hospitalization only; loop diuretics give symptom relief without mortality benefit
Q4. He received maximum Furosemide and a thiazide without improvement: what can you add?	• **Acetazolamide 500 mg IV daily** (ADVOR) or add Spironolactone 25–50 mg (sequential nephron blockade); switch to Bumetanide/Torsemide or continuous Furosemide infusion • Correct hypoalbuminemia/hyponatremia, stop NSAIDs; ultrafiltration or CRRT if refractory; Dobutamine/Milrinone only if low-output (cold) with hypoperfusion
Q5. How do you monitor Nitroglycerin?	• BP (hold if SBP <90–100), headache, tachycardia, symptom relief, urine output; tolerance after 24–48 h (needs dose escalation); high doses: methemoglobinemia • A pulmonary-artery (Swan-Ganz) catheter is not routine: reserved for unclear volume status or shock
Q6. Which anticoagulant can you use?	• **Heparin:** UFH 5000 units SC q8–12h or Enoxaparin 40 mg daily for VTE prophylaxis; for AF stroke prevention CHA2DS2-VASc ≥2: UFH infusion bridging after surgery when bleeding risk is acceptable, then a DOAC (Apixaban 5 mg BID)
Q7. What is the AF plan? Mention rate and rhythm control.	• Rate: β-blocker once euvolemic (Metoprolol succinate, Bisoprolol), **Digoxin** 0.125–0.25 mg daily (level 0.5–0.9 ng/mL), IV **Amiodarone** (150 mg bolus, then 1 mg/min × 6 h, then 0.5 mg/min × 18 h) • Rhythm: **Amiodarone or Dofetilide** (HFrEF); electrical cardioversion if unstable; ablation in HFrEF (CASTLE-AF) • Avoid non-DHP CCB, Flecainide, Propafenone, Dronedarone and Sotalol in HFrEF
Q8. Depression: which antidepressant class?	• **SSRI: Sertraline** 25–50 mg daily (safe in HF: SADHART-CHF); avoid tricyclics, high-dose Citalopram (QT) and Venlafaxine (BP)
Q9. Is he a candidate for TPN?	• **Yes:** NPO >7 days after GI surgery and the enteral route is not usable (ASPEN: PN after day 7 if low nutrition risk, sooner if high risk); use a concentrated formula (fluid restriction) and correct K, Mg, Phos first (refeeding)
Q10. Calculate total calories at 25 kcal/kg/day.	• 80 × 25 = **2000 kcal/day**
Q11. Calculate the calories from Dextrose at 4 g/kg/day.	• 4 × 80 = **320 g** × 3.4 kcal/g = **1088 kcal**; GIR = 320,000 ÷ 80 ÷ 1440 = 2.8 mg/kg/min (limit 4–5)
Also be ready for (exam add-ons)	• Goals: relieve congestion and dyspnea, restore perfusion, identify and treat the trigger (infection, ischemia, arrhythmia, non-adherence), then optimize GDMT before discharge • Non-pharm: sodium restriction, fluid restriction if hyponatremic, daily weights and input/output, oxygen if SpO2 <90%, upright position, early mobilization, VTE prophylaxis • First-line: Furosemide IV 20–40 mg (2–2.5× the home oral dose), reassess urine output >100–150 mL/h at 2 h and sodium at 6 h; add Metolazone 2.5–5 mg or Acetazolamide for resistance; continue GDMT

Question	Answer
	unless shock or K >5 • Pregnancy: avoid ACE inhibitor/ARB/ARNI/MRA/SGLT2 inhibitors; Furosemide, Hydralazine + nitrate, Metoprolol are used; check for peripartum cardiomyopathy • Renal/dialysis: cardiorenal syndrome: continue decongestion; monitor SCr, K, Mg; adjust antibiotics (Levofloxacin 750 mg q48h if CrCl 20–49); stop Metformin if eGFR <30 or unstable • Liver: congestive hepatopathy; Spironolactone and Furosemide ratio 100:40; avoid hepatotoxic drugs • Infection (UTI/pyelonephritis): choose a non-penicillin agent for severe allergy (Levofloxacin 750 mg daily × 5 days; Ceftriaxone is acceptable if the reaction was not severe); avoid QT stacking • Counseling: daily weights, sodium and fluid advice, medication adherence, when to call, Entresto 36-hour washout, hyperkalemia (K 5.6: hold K-sparing agents and recheck)

Case 14: AF in a 57-kg Woman, then STEMI ★ Repeated case, also in 2022–2023, 2021, 2018–2019

Case details: 69-year-old woman with HTN and DM on Warfarin 6 mg (INR 2.5), weight 57 kg, CrCl 95 mL/min. Also Diltiazem 160 mg daily, Flecainide 50 mg BID, Lisinopril 20 mg daily, Metformin 500 mg BID. HR 95 with fatigue.

Question	Answer
Q1. What are her CHA2DS2-VASc and risk factors?	• HTN 1 + DM 1 + age 65–74 1 + female sex 1 = **4** (HF 0, stroke 0, vascular 0); after the STEMI add vascular disease: **5**
Q2. What is her HAS-BLED?	• Age >65: 1; uncontrolled HTN (SBP >160) not given: 0; labile INR, bleeding history, alcohol, antiplatelets/NSAIDs: 0 → **1** (missing data: state your assumption)
Q3. HR 95 but fatigued: what will you do? Which dose will you increase, and to what target?	• Symptomatic AF: strict rate control, **resting HR <80 bpm** (lenient <110 is for asymptomatic patients) • **Increase Diltiazem** (160 → 240 mg daily; max 360 mg) if EF is preserved, or add/switch to a β-blocker; check TSH, anemia, volume status
Q4. She has a STEMI treated successfully: what changes in her AF medicines?	• **Stop Flecainide** (contraindicated after MI/structural disease: CAST) → Amiodarone (or Dofetilide/Sotalol) • **Replace Diltiazem with a β-blocker** (post-MI benefit, avoid Diltiazem if EF reduced) • Antithrombotic after PCI in AF: OAC + Clopidogrel, with Aspirin only 1–4 weeks (2025 ACC/AHA), then OAC alone after 12 months; high-intensity statin
Q5. She cannot monitor INR: switching to a DOAC. In general, what are the apixaban dose-adjustment criteria?	• Reduce to **2.5 mg BID if any 2 of:** age ≥80 years, weight ≤60 kg, SCr ≥1.5 mg/dL
Q6. Does she meet the criteria? Which dose?	• Weight 57 kg (≤60) = **only 1 criterion**; age 69 and normal SCr (CrCl 95): **Apixaban 5 mg BID** (no reduction)
Q7. How do you switch from Warfarin to Apixaban? Mention all steps.	• Stop Warfarin, check INR; start Apixaban when **INR <2.0** (if INR is 2.0–2.5, recheck the next day and start once <2.0); no bridging • Counsel: twice-daily adherence, no INR monitoring, bleeding signs, missed-dose rules, check interactions (Diltiazem raises levels modestly: no change), CBC/SCr/LFT yearly
Also be ready for (exam add-ons)	• Goals: prevent stroke (anticoagulation), control rate or rhythm, treat the cause, improve symptoms and prevent HF/tachycardiomyopathy • Non-pharm: weight loss, alcohol reduction, treat sleep apnea, exercise, BP control, smoking cessation, avoid stimulants (caffeine, decongestants); catheter ablation if drug failure • First-line: anticoagulant by CHA2DS2-VASc (men ≥2, women ≥3): Apixaban 5 mg BID (2.5 mg if 2 of age ≥80, weight ≤60 kg, SCr ≥1.5), Rivaroxaban 20 mg with the evening meal (15 mg if CrCl 15–50), or Warfarin INR 2–3 (valvular AF); rate control: Metoprolol/Bisoprolol or Diltiazem (not in HFrEF); rhythm: Flecainide/Propafenone (no structural disease), Amiodarone/Dofetilide (HF, CAD) • Pregnancy: DOACs and Warfarin contraindicated; use LMWH (Enoxaparin 1 mg/kg q12h) or UFH; rate control with Metoprolol or Labetalol; cardioversion is safe; avoid Amiodarone, Dronedarone • Renal/dialysis: Apixaban 5 mg BID (2.5 mg if age ≥80 or weight ≤60) in HD; Dabigatran avoid if CrCl <30; Rivaroxaban 15 mg if CrCl 15–50, avoid <15; Edoxaban 30 mg if CrCl 15–50; Dofetilide/Sotalol adjust to CrCl • Liver: avoid DOACs in Child-Pugh C (Rivaroxaban avoided in B and C); Amiodarone monitor LFTs; Warfarin baseline INR is high • HF: avoid Flecainide, Propafenone, Dronedarone, Sotalol and Diltiazem/Verapamil in HFrEF; use Amiodarone or Dofetilide; β-blocker for rate • GI (PUD/GERD): PPI when combined with antiplatelets; bleeding risk; HAS-BLED corrections (stop NSAIDs, control BP) • Counseling: adherence (DOAC half-life is short), bleeding signs, Rivaroxaban with food, Warfarin: steady Vitamin K intake and INR checks, drug interactions (Amiodarone, Clarithromycin, Rifampin, Carbamazepine), missed dose rules, show the card to all clinicians • Side scenario (post-MI): stop Flecainide/Propafenone (CAST) and Diltiazem if EF reduced; use a β-blocker, Amiodarone or Dofetilide; with PCI use Apixaban + Clopidogrel (stop Aspirin after 1 week to 1 month), then anticoagulant alone at 12 months; high-intensity statin

Case 15: STEMI in a Patient with Depression on Venlafaxine ★ Repeated case, also in 2025, 2022–2023

Case details: Male smoker with HTN and dyslipidemia, LDL 5 mmol/L (193 mg/dL), depression on Venlafaxine 75 mg daily, Bisoprolol 2.5 mg and Enalapril 10 mg daily. Chest pain; the hospital has a cath lab. ECG shows STEMI, troponin high; he took Nitroglycerin and Morphine.

Question	Answer
Q1. PCI or fibrinolytics? Justify. Short- and long-term benefits? Major risk of fibrinolytics?	• **Primary PCI** (cath lab available, first medical contact to device ≤90 min; ≤120 min if transferred): higher infarct-artery patency, lower mortality, reinfarction and stroke • Short-term: restores TIMI 3 flow, less recurrent MI, shock, arrhythmia, mechanical complications; long term: better survival, less HF • Major fibrinolytic risk: **intracranial hemorrhage (about 1%)** and major bleeding
Q2. What are the benefits of PCI over fibrinolytics (short-term complications of STEMI)?	• Avoids failed reperfusion (about 30% with lytics), reinfarction, intracranial hemorrhage and early mechanical complications (VSD, papillary-muscle rupture, free-wall rupture), cardiogenic shock and VF
Q3. He will go to PCI: what will you give?	• Aspirin 162–325 mg chewed; P2Y12 loading: **Ticagrelor 180 mg** (preferred; or Prasugrel 60 mg; Clopidogrel 600 mg); UFH 70–100 units/kg IV bolus (50–70 with a GP IIb/IIIa inhibitor); oxygen only if SpO2 <90% • Atorvastatin 80 mg early; opioids (Morphine) delay oral P2Y12 absorption: use Fentanyl or crush Ticagrelor
Q4. What do you want to change in his medicines, and why?	• **Statin:** start Atorvastatin 80 mg (LDL 193 mg/dL; goal <55 mg/dL; add Ezetimibe if ≥70 on maximal statin) • Continue Enalapril (post-MI with HTN) and the β-blocker (Bisoprolol: titrate; reassess at 1 year if LVEF is preserved); stop smoking (Varenicline or NRT) • **Change Venlafaxine** (see Q5)
Q5. Will you change Venlafaxine? Why? How will you discontinue it?	• Yes: Venlafaxine raises BP and HR (dose-related), has less cardiac-safety evidence, and may prolong QT; **Sertraline** is preferred after MI (SADHART) • Do **not stop abruptly** (discontinuation syndrome): taper 75 mg → 37.5 mg for 1–2 weeks → stop, while starting Sertraline 25 mg → 50 mg; monitor dizziness, nausea, anxiety, and serotonin-toxicity signs during overlap
Q6. Which class for his depression? Name 3 medicines.	• **SSRIs:** Sertraline 50–200 mg, Escitalopram 10–20 mg (QT dose-related), Fluoxetine 20–40 mg (or Paroxetine, Citalopram ≤40 mg); bleeding risk rises with DAPT (add a PPI)
Q7. List all discharge medicines.	• Aspirin 81 mg daily (indefinite); Ticagrelor 90 mg BID (or Clopidogrel 75 mg) for 12 months; Atorvastatin 80 mg; Bisoprolol (titrate); Enalapril 10–20 mg BID; Sertraline; Pantoprazole 40 mg (GI protection with DAPT + SSRI); smoking-cessation therapy • **Nitroglycerin 0.4 mg SL PRN** every 5 min × 3 doses; call emergency services if pain persists 5 min after the first dose • Cardiac rehabilitation, vaccines (influenza), lifestyle
Also be ready for (exam add-ons)	• Goals: restore infarct-artery patency quickly (door-to-balloon ≤90 min, door-to-needle ≤30 min), limit infarct size, relieve ischemia, prevent arrhythmia, heart failure and reinfarction • Non-pharm: primary PCI (preferred), cardiac rehab, smoking cessation, Mediterranean diet, exercise 150 min/week, weight control, sodium restriction, vaccines (influenza), driving advice • First-line: Aspirin 162–325 mg chewed, P2Y12 inhibitor (Ticagrelor 180 mg or Prasugrel 60 mg or Clopidogrel 600 mg), UFH 70–100 units/kg in the cath lab, Atorvastatin 80 mg, oral β-blocker within 24 h if LVEF reduced, HF or ventricular arrhythmia and no shock (2025 ACC/AHA; reassess after 1 year if LVEF is preserved), Nitroglycerin SL for pain • Pregnancy: MI in pregnancy is rare: PCI preferred, avoid fibrinolysis if possible; low-dose Aspirin and Clopidogrel acceptable; avoid ACE inhibitors/ARBs/MRA and statins; Metoprolol and Labetalol are safe; UFH or Enoxaparin acceptable • Renal/dialysis: contrast nephropathy prevention (hydration, minimal contrast, hold Metformin and nephrotoxins); Enoxaparin 1 mg/kg q24h if CrCl <30; Bivalirudin: reduce the infusion if CrCl <30; Ticagrelor needs no adjustment • Liver: avoid Ticagrelor in severe hepatic impairment; statin caution when ALT >3× ULN; bleeding risk higher • HF/shock: avoid β-blocker in acute HF or shock; add Spironolactone if EF <40% with DM or HF; ACE inhibitor/ARNI for EF <40% • GI (PUD/GERD): add PPI with DAPT in patients at GI-bleeding risk (Pantoprazole; avoid Omeprazole with Clopidogrel) • LDL-C goal <55 mg/dL (add Ezetimibe if ≥70 on maximal statin, 2025 ACC/AHA); Counseling: DAPT adherence for 12 months (never stop early), bleeding signs, Ticagrelor dyspnea and Aspirin dose ≤100 mg, Nitroglycerin use (call 911 if pain persists 5 min after first dose), statin muscle pain, cardiac rehab, return to activity • Depression: SSRI (Sertraline) is preferred after MI; avoid TCAs, high-dose Citalopram and Venlafaxine (BP/QT); SSRIs + DAPT raise bleeding risk (add a PPI); taper SNRIs to avoid discontinuation syndrome

2022–2023

Case 1: HFrEF (NYHA III)

Case details: HF NYHA class III, LVEF 30%, LDL-C 100 mg/dL. Home: Enalapril 10 mg BID, Diltiazem 60 mg once, Atorvastatin 40 mg (cannot increase because of adverse effects).

Question	Answer
Q1. What is the best management for his HF?	• Stop Diltiazem: non-DHP CCB is harmful in HFrEF • Switch Enalapril to ARNI: stop Enalapril, **36-hour washout**, then Sacubitril/valsartan ◦ Enalapril 20 mg/day is a standard dose (>10 mg/day) → start **49/51 mg BID**, titrate to 97/103 mg BID • Add: evidence-based beta-blocker, MRA, SGLT2i • Doses: ◦ Bisoprolol 1.25 → 10 mg daily, Carvedilol 3.125 → 25 mg BID, or Metoprolol succinate 12.5–25 → 200 mg daily ◦ Spironolactone 12.5–25 mg daily (eGFR >30, K <5.0) ◦ Dapagliflozin or Empagliflozin 10 mg daily • Monitor: BP, HR, SCr and K 1–2 weeks after each start/titration; weight and congestion each visit • Monitor MRA: K and SCr ~1 week, 1 month, 6 months; halve or stop if K >5.5 • Non-pharm: sodium <2300 mg/day, daily weights, exercise/cardiac rehab, vaccines, avoid NSAIDs and non-DHP CCBs
Q2. What do you do with his dyslipidemia?	• Clarify the indication: is the HF ischemic (ASCVD)? • If ASCVD: on maximally tolerated statin with LDL-C ≥70 → add Ezetimibe 10 mg daily • If primary prevention: Atorvastatin 40 mg is already high intensity; no change needed
Q3. Doctor wants to shift from Enalapril to Losartan. Do you agree? Why?	• No: ARBs are not superior to ACE inhibitors in HFrEF • Switch to ARB only if ACE inhibitor cough or intolerance • Better move: switch to ARNI
Q4. Role of beta-blockers in this patient? All the benefits?	• Mortality: about 35% relative reduction (including sudden cardiac death) • Hospitalizations: about 25% fewer • Symptoms and clinical status: improve over months • Remodeling: reverse LV remodeling, raise EF • Agents: Bisoprolol, Carvedilol, Metoprolol succinate only
Q5. Do you recommend adding digitalis? Why?	• Not now: GDMT is not optimized • Digoxin reduces HF hospitalization only, not mortality (DIG trial) • Reserve for: persistent symptoms despite optimized GDMT, or AF rate control • Target level: 0.5–0.9 ng/mL
Q6. Mention all classes that reduce mortality in HF.	• ARNI / ACE inhibitor / ARB • Evidence-based beta-blockers • MRA • SGLT2 inhibitors • Hydralazine/ISDN (African American patients, NYHA III–IV) • Not mortality-reducing: ivabradine, vericiguat, Digoxin (reduce hospitalization) • Devices: ICD and CRT
Also be ready for (exam add-ons)	• Goals: reduce mortality and HF hospitalization, relieve congestion and symptoms, prevent remodeling, improve exercise tolerance and quality of life • Non-pharm: sodium <2–3 g/day, fluid 1.5–2 L only if hyponatremia or advanced HF, daily weights, exercise or cardiac rehabilitation, vaccines (influenza, pneumococcal, COVID-19), stop smoking, limit alcohol, avoid NSAIDs • First-line: 4 pillars: ARNI Sacubitril/valsartan 24/26 mg BID (or ACE inhibitor/ARB), Bisoprolol 1.25 mg (or Carvedilol 3.125 mg BID, Metoprolol succinate 12.5–25 mg) when euvolemic, Spironolactone 12.5–25 mg (eGFR >30, K <5), Dapagliflozin 10 mg or Empagliflozin 10 mg; Furosemide 20–40 mg only for congestion • Pregnancy: ACE inhibitor, ARB, ARNI, MRA and SGLT2 inhibitors are contraindicated; use Metoprolol, Labetalol, Hydralazine + Isosorbide dinitrate and a loop diuretic; peripartum cardiomyopathy: add Bromocriptine and anticoagulation per specialist; counsel contraception • Renal/dialysis: check SCr and K (ARNI, ACE inhibitor, MRA); avoid MRA if eGFR <30 or K >5; Empagliflozin/Dapagliflozin can start at eGFR ≥20; higher loop-diuretic doses needed; Digoxin reduce in CKD • Liver: avoid ARNI in Child-Pugh C; Carvedilol avoid in severe disease; congestive hepatopathy improves with decongestion • GI (PUD/GERD): if on antiplatelet or anticoagulant add a PPI (Pantoprazole preferred with Clopidogrel); avoid NSAIDs • Counseling: ARNI: dizziness, angioedema, hyperkalemia, 36-hour ACE-inhibitor washout; β-blocker: fatigue, do not stop abruptly; Spironolactone: avoid potassium salt substitutes, gynecomastia; SGLT2 inhibitor: genital infection, hydration, hold on sick days; weigh daily and call if gain >1 kg in a day or 2 kg in a week

Case 2: STEMI ★ Repeated case, also in 2025, 2024, 2018–2019

Case details: Radiating chest pain; diagnosed STEMI.

Question	Answer
Q1. What is the goal of treatment in general?	• Restore patency of the infarct-related artery and minimize infarct size • Prevent complications (arrhythmia, HF, death) • Control chest pain and associated symptoms
Q2. Drugs for the acute phase in the emergency department, with doses?	• Strategy: primary PCI • Aspirin: 162–325 mg chewed • Nitroglycerin: 0.3–0.4 mg SL/spray q5min up to 3 doses • Morphine: 2–4 mg IV q5–15 min only if pain persists • Oxygen if SaO2 <90% • Oral beta-blocker within 24 h if no contraindication • P2Y12 load: Ticagrelor 180 mg or Prasugrel 60 mg (or Clopidogrel 600 mg) • Anticoagulant: UFH 70–100 units/kg IV bolus during PCI • High-intensity statin • Monitor: ECG, BP, HR, SaO2, troponin, Hgb, SCr, ACT during PCI
Q3. SCr rose from 190 to 197 (AKI) 2 days after PCI. What is the cause?	• Contrast-associated AKI: SCr rises within 24 h and peaks 2–5 days after contrast • Also exclude: hypotension, atheroemboli, nephrotoxic drugs
Q4. Management of his AKI?	• Supportive: isotonic fluids if volume depleted • Stop nephrotoxins: NSAIDs; hold ACE inhibitor/ARB/MRA and Metformin until recovery • Renally adjust drugs • Monitor: SCr, UOP, K, acid–base • Dialysis only for standard indications
Q5. Chronic management for his STEMI?	• DAPT: Aspirin 81 mg + Ticagrelor 90 mg BID (preferred) or Clopidogrel 75 mg, ≥12 months • Beta-blocker: Metoprolol succinate, Carvedilol or Bisoprolol • ACE inhibitor: if EF ≤40%, HTN, DM, CKD • High-intensity statin ± Ezetimibe: LDL-C <55 mg/dL (1.4 mmol/L) reasonable • MRA: if on beta-blocker and ACE inhibitor with EF ≤40% and HF or DM • SL Nitroglycerin 0.4 mg PRN • PPI with DAPT if high GI bleeding risk • Cardiac rehabilitation • Doses: ◦ Aspirin 81 mg daily (indefinitely) ◦ Ticagrelor 90 mg BID or Clopidogrel 75 mg daily for 12 months ◦ Atorvastatin 80 mg daily ± Ezetimibe 10 mg ◦ Metoprolol succinate 25 → 200 mg daily or Bisoprolol 1.25 → 10 mg daily ◦ Lisinopril 2.5–5 → 20–40 mg daily ◦ Nitroglycerin 0.4 mg SL q5min × 3 PRN chest pain ◦ Eplerenone 25 → 50 mg daily (if MRA indicated) • Monitor: chest pain, bleeding (Hgb), BP, HR, SCr/K (ACE inhibitor), lipids 4–8 weeks, LFT baseline, adherence to DAPT • Non-pharm: smoking cessation, diet, exercise, weight, cardiac rehab
Q6. After 2 months, potassium rose from 4 to 5. What to do?	• K 5.0 is acceptable: continue therapy and recheck • K >5.5: halve the MRA dose or stop • Review: K supplements, salt substitutes, NSAIDs, diet
Also be ready for (exam add-ons)	• Goals: restore infarct-artery patency quickly (door-to-balloon ≤90 min, door-to-needle ≤30 min), limit infarct size, relieve ischemia, prevent arrhythmia, heart failure and reinfarction • Non-pharm: primary PCI (preferred), cardiac rehab, smoking cessation, Mediterranean diet, exercise 150 min/week, weight control, sodium restriction, vaccines (influenza), driving advice • First-line: Aspirin 162–325 mg chewed, P2Y12 inhibitor (Ticagrelor 180 mg or Prasugrel 60 mg or Clopidogrel 600 mg), UFH 70–100 units/kg in the cath lab, Atorvastatin 80 mg, oral β-blocker within 24 h if LVEF reduced, HF or ventricular arrhythmia and no shock (2025 ACC/AHA; reassess after 1 year if LVEF is preserved), Nitroglycerin SL for pain • Pregnancy: MI in pregnancy is rare: PCI preferred, avoid fibrinolysis if possible; low-dose Aspirin and Clopidogrel acceptable; avoid ACE inhibitors/ARBs/MRA and statins; Metoprolol and Labetalol are safe; UFH or Enoxaparin acceptable • Renal/dialysis: contrast nephropathy prevention (hydration, minimal contrast, hold Metformin and nephrotoxins); Enoxaparin 1 mg/kg q24h if CrCl <30; Bivalirudin: reduce the infusion if CrCl <30; Ticagrelor needs no adjustment • Liver: avoid Ticagrelor in severe hepatic impairment; statin caution when ALT >3× ULN; bleeding risk higher • HF/shock: avoid β-blocker in acute HF or shock; add Spironolactone if EF <40% with DM or HF; ACE inhibitor/ARNI for EF <40% • GI (PUD/GERD): add PPI with DAPT in patients at GI-bleeding risk (Pantoprazole; avoid Omeprazole with Clopidogrel)

Question	Answer
	• LDL-C goal <55 mg/dL (add Ezetimibe if ≥70 on maximal statin, 2025 ACC/AHA); Counseling: DAPT adherence for 12 months (never stop early), bleeding signs, Ticagrelor dyspnea and Aspirin dose ≤100 mg, Nitroglycerin use (call 911 if pain persists 5 min after first dose), statin muscle pain, cardiac rehab, return to activity

Case 3: ADHF (with UTI)

Case details: Known DM, HTN, HFrEF. SOB, CXR pulmonary edema, +3 pitting edema. Allergy: penicillin (severe), peanut. Home: Furosemide 20 mg PO OD, Lisinopril 10 mg PO BID, Atorvastatin 80 mg OD, Metoprolol 100 mg OD, Metformin. K 5.6.

Question	Answer
Q1. What is the diagnosis?	• ADHF (warm and wet): pulmonary edema, peripheral edema, dyspnea
Q2. Causes of ADHF (all causes)?	• Medication related: non-adherence; negative inotropes (non-DHP CCB, class I antiarrhythmics); sodium/fluid retainers (NSAIDs, TZDs, steroids); alcohol or illicit drugs • Patient/disease related: dietary sodium or fluid non-adherence; acute ischemia; uncontrolled HTN; AF or other arrhythmias; infection; PE; thyroid disease; anemia; worsening renal function
Q3. What do you recommend for edema (agent, dose, frequency)? If it fails?	• IV Furosemide: home 20 mg PO = 10 mg IV; give 1–2.5× home dose IV, eg 20–40 mg IV BID • Double the dose until response (UOP >500 mL in 2 hours) • If still inadequate: add Metolazone 2.5–5 mg PO 30 min before the loop, or Chlorothiazide 250–500 mg IV • K 5.6: hold Lisinopril for now; no MRA until K <5.0 • Monitor: UOP hourly, intake/output, daily weight, SCr, K, Mg, Na every 12–24 h, BP, symptoms
Q4. New dysuria, flank pain, fever, leukocytosis. Diagnosis? Type of UTI?	• Acute pyelonephritis = complicated UTI (upper tract with systemic signs)
Q5. Treatment of his UTI, and for how long (severe penicillin allergy)?	• Levofloxacin 750 mg IV/PO q24h × 5–7 days (if local FQ resistance ≤10%); adjust for CrCl • Alternative: Ciprofloxacin 400 mg IV q12h; Aztreonam if FQ not suitable • Tailor to culture • Monitor: fever and symptoms at 48–72 h, culture and susceptibility, QTc, SCr, glucose
Q6. What will you modify in his home medications (K now normal, euvolemic)?	• Switch Lisinopril (20 mg/day) to Sacubitril/valsartan 49/51 mg BID after a **36-hour washout** • Beta-blocker: change to Metoprolol succinate and titrate • Add Spironolactone 12.5–25 mg daily (eGFR >30, K <5.0) • Add SGLT2i: Empagliflozin or Dapagliflozin 10 mg daily (also for DM) • Furosemide: higher oral maintenance dose to keep euvolemia • Metformin: continue if eGFR ≥30 • Monitor: SCr and K 1–2 weeks after ARNI/MRA, BP, weight, glucose
Q7. Discharge medications (including the antibiotic)?	• Sacubitril/valsartan 49/51 mg BID • Metoprolol succinate 100 mg daily, titrate to 200 mg • Spironolactone 12.5–25 mg daily • Empagliflozin 10 mg daily • Furosemide at the new oral maintenance dose • Atorvastatin 80 mg daily • Metformin if renal function allows • Levofloxacin 750 mg PO daily to complete 5–7 days total • Counsel: daily weights, sodium limit, K and SCr in 1 week
Also be ready for (exam add-ons)	• Goals: relieve congestion and dyspnea, restore perfusion, identify and treat the trigger (infection, ischemia, arrhythmia, non-adherence), then optimize GDMT before discharge • Non-pharm: sodium restriction, fluid restriction if hyponatremic, daily weights and input/output, oxygen if SpO2 <90%, upright position, early mobilization, VTE prophylaxis • First-line: Furosemide IV 20–40 mg (2–2.5× the home oral dose), reassess urine output >100–150 mL/h at 2 h and sodium at 6 h; add Metolazone 2.5–5 mg or Acetazolamide for resistance; continue GDMT unless shock or K >5 • Pregnancy: avoid ACE inhibitor/ARB/ARNI/MRA/SGLT2 inhibitors; Furosemide, Hydralazine + nitrate, Metoprolol are used; check for peripartum cardiomyopathy • Renal/dialysis: cardiorenal syndrome: continue decongestion; monitor SCr, K, Mg; adjust antibiotics (Levofloxacin 750 mg q48h if CrCl 20–49); stop Metformin if eGFR <30 or unstable • Liver: congestive hepatopathy; Spironolactone and Furosemide ratio 100:40; avoid hepatotoxic drugs • Infection (UTI/pyelonephritis): choose a non-penicillin agent for severe allergy (Levofloxacin 750 mg daily × 5 days; Ceftriaxone is acceptable if the reaction was not severe); avoid QT stacking • Counseling: daily weights, sodium and fluid advice, medication adherence, when to call, Entresto 36-hour washout, hyperkalemia (K 5.6: hold K-sparing agents and recheck)

Case 4: HFrEF Stage C

Case details: HFrEF, stage C.

Question	Answer
Q1. HFrEF (EF <40%) with arrhythmia: how will you treat it?	• AF rate control: beta-blocker ± Digoxin; avoid non-DHP CCB • AF rhythm control: Amiodarone or Dofetilide; consider catheter ablation • Avoid: class Ic, Dronedarone (NYHA III–IV or recent decompensation) • Ventricular arrhythmia: ICD; Amiodarone as adjunct • Doses: ◦ Metoprolol succinate 25 → 200 mg daily; Digoxin 0.125 mg daily (level 0.5–0.9) ◦ Amiodarone 400 mg BID–TID to 6–10 g, then 200 mg daily ◦ Dofetilide 500 mcg BID if CrCl >60 (start in hospital) • Monitor: HR, ECG/QTc, K, Mg, SCr, TFT/LFT with Amiodarone, Digoxin level
Q2. Lifelong medications after MI?	• Beta-blocker • ACE inhibitor/ARB/ARNI • High-intensity statin ± Ezetimibe • Aspirin (after 12 months of DAPT) • MRA and SGLT2i with HFrEF • Doses: ◦ Aspirin 81 mg daily (indefinitely) ◦ Ticagrelor 90 mg BID or Clopidogrel 75 mg daily for 12 months ◦ Atorvastatin 80 mg daily ± Ezetimibe 10 mg ◦ Metoprolol succinate 25 → 200 mg daily or Bisoprolol 1.25 → 10 mg daily ◦ Lisinopril 2.5–5 → 20–40 mg daily ◦ Nitroglycerin 0.4 mg SL q5min × 3 PRN chest pain • Monitor: chest pain, bleeding (Hgb), BP, HR, SCr/K (ACE inhibitor), lipids 4–8 weeks, LFT baseline, adherence to DAPT
Also be ready for (exam add-ons)	• Goals: reduce mortality and HF hospitalization, relieve congestion and symptoms, prevent remodeling, improve exercise tolerance and quality of life • Non-pharm: sodium <2–3 g/day, fluid 1.5–2 L only if hyponatremia or advanced HF, daily weights, exercise or cardiac rehabilitation, vaccines (influenza, pneumococcal, COVID-19), stop smoking, limit alcohol, avoid NSAIDs • First-line: 4 pillars: ARNI Sacubitril/valsartan 24/26 mg BID (or ACE inhibitor/ARB), Bisoprolol 1.25 mg (or Carvedilol 3.125 mg BID, Metoprolol succinate 12.5–25 mg) when euvolemic, Spironolactone 12.5–25 mg (eGFR >30, K <5), Dapagliflozin 10 mg or Empagliflozin 10 mg; Furosemide 20–40 mg only for congestion • Pregnancy: ACE inhibitor, ARB, ARNI, MRA and SGLT2 inhibitors are contraindicated; use Metoprolol, Labetalol, Hydralazine + Isosorbide dinitrate and a loop diuretic; peripartum cardiomyopathy: add Bromocriptine and anticoagulation per specialist; counsel contraception • Renal/dialysis: check SCr and K (ARNI, ACE inhibitor, MRA); avoid MRA if eGFR <30 or K >5; Empagliflozin/Dapagliflozin can start at eGFR ≥20; higher loop-diuretic doses needed; Digoxin reduce in CKD • Liver: avoid ARNI in Child-Pugh C; Carvedilol avoid in severe disease; congestive hepatopathy improves with decongestion • GI (PUD/GERD): if on antiplatelet or anticoagulant add a PPI (Pantoprazole preferred with Clopidogrel); avoid NSAIDs • Counseling: ARNI: dizziness, angioedema, hyperkalemia, 36-hour ACE-inhibitor washout; β-blocker: fatigue, do not stop abruptly; Spironolactone: avoid potassium salt substitutes, gynecomastia; SGLT2 inhibitor: genital infection, hydration, hold on sick days; weigh daily and call if gain >1 kg in a day or 2 kg in a week

Case 5: Atrial Fibrillation (SOE)

Case details: Khaled, 55-year-old man, 75 kg, BMI 32, HTN and T2DM, new AF. CHA2DS2-VASc annual stroke risk 2.1%. SCr 1.3 mg/dL, CrCl 43 mL/min (actual body weight), INR 0.9, BP 126/78, A1C 7.2%.

Question	Answer
Q1. Risk factors for stroke in this patient?	• In the score: HTN, DM • Other: AF itself, obesity, male sex • Age 55 does not score (points start at 65)
Q2. Appropriate treatment strategy?	• Anticoagulation: CHA2DS2-VASc = 2 (HTN + DM) in a man → OAC recommended • Rate or rhythm control by symptoms • Risk-factor modification: weight loss, BP, glucose, sleep apnea, alcohol • OAC dose: see next question (Rivaroxaban 15 mg daily) • Rate control: Metoprolol succinate 25–200 mg daily or Diltiazem ER 120–360 mg daily (no HFrEF) • Monitor: HR, BP, symptoms, SCr/CBC yearly for DOAC, bleeding
Q3. Appropriate once-daily DOAC based on CrCl?	• Rivaroxaban **15 mg daily with the evening meal** (CrCl 15–50) • or Edoxaban 30 mg daily (CrCl 15–50) • Note: DOAC trials used Cockcroft-Gault with actual body weight

Question	Answer
Q4. Appropriate antiarrhythmic for cardioversion?	• No structural heart disease: Flecainide or Propafenone first line ◦ Flecainide 300 mg PO once (pill-in-the-pocket), with an AV nodal blocker • Others: Amiodarone, Ibutilide, Dofetilide • Before cardioversion: ≥3 weeks anticoagulation or TEE if AF ≥48 h; continue ≥4 weeks after
Q5. Target heart rate?	• Lenient: resting HR <100–110 if asymptomatic
Q6. If he develops HFrEF, appropriate antiarrhythmic maintenance?	• Amiodarone or Dofetilide • Catheter ablation is also recommended in HFrEF
Q7. Admitted to ICU and starting dialysis. Would you change the DOAC?	• Yes, switch to Apixaban **5 mg BID** ◦ 2.5 mg BID only if age ≥80 or weight ≤60 kg (he is 55 y, 75 kg) • Rivaroxaban is labeled 15 mg daily in dialysis but supported only by small PK data • Avoid Edoxaban and Dabigatran in dialysis
Also be ready for (exam add-ons)	• Goals: prevent stroke (anticoagulation), control rate or rhythm, treat the cause, improve symptoms and prevent HF/tachycardiomyopathy • Non-pharm: weight loss, alcohol reduction, treat sleep apnea, exercise, BP control, smoking cessation, avoid stimulants (caffeine, decongestants); catheter ablation if drug failure • First-line: anticoagulant by CHA2DS2-VASc (men ≥2, women ≥3): Apixaban 5 mg BID (2.5 mg if 2 of age ≥80, weight ≤60 kg, SCr ≥1.5), Rivaroxaban 20 mg with the evening meal (15 mg if CrCl 15–50), or Warfarin INR 2–3 (valvular AF); rate control: Metoprolol/Bisoprolol or Diltiazem (not in HFrEF); rhythm: Flecainide/Propafenone (no structural disease), Amiodarone/Dofetilide (HF, CAD) • Pregnancy: DOACs and Warfarin contraindicated; use LMWH (Enoxaparin 1 mg/kg q12h) or UFH; rate control with Metoprolol or Labetalol; cardioversion is safe; avoid Amiodarone, Dronedarone • Renal/dialysis: Apixaban 5 mg BID (2.5 mg if age ≥80 or weight ≤60) in HD; Dabigatran avoid if CrCl <30; Rivaroxaban 15 mg if CrCl 15–50, avoid <15; Edoxaban 30 mg if CrCl 15–50; Dofetilide/Sotalol adjust to CrCl • Liver: avoid DOACs in Child-Pugh C (Rivaroxaban avoided in B and C); Amiodarone monitor LFTs; Warfarin baseline INR is high • HF: avoid Flecainide, Propafenone, Dronedarone, Sotalol and Diltiazem/Verapamil in HFrEF; use Amiodarone or Dofetilide; β-blocker for rate • GI (PUD/GERD): PPI when combined with antiplatelets; bleeding risk; HAS-BLED corrections (stop NSAIDs, control BP) • Counseling: adherence (DOAC half-life is short), bleeding signs, Rivaroxaban with food, Warfarin: steady Vitamin K intake and INR checks, drug interactions (Amiodarone, Clarithromycin, Rifampin, Carbamazepine), missed dose rules, show the card to all clinicians

Case 6: PE in a Young Patient ★ Repeated case, also in 2024

Case details: 21-year-old woman, medically free; sister had DVT/PE. Chest pain and dyspnea; V/Q scan shows PE. INR 0.9, Hgb normal. Thrombophilia and antiphospholipid tests pending.

Question	Answer
Q1. What confirms her diagnosis of PE?	• Symptoms are non-specific; diagnosis is confirmed by imaging: V/Q scan (or CT pulmonary angiography) • Pretest probability: Wells score guides D-dimer vs direct imaging
Q2. Why was a hypercoagulability test done?	• Young, positive family history, unprovoked VTE • Looks for: inherited (factor V Leiden, prothrombin G20210A, protein C/S and antithrombin deficiency) and acquired (antiphospholipid syndrome) • Changes management: APS → Warfarin preferred over DOAC; may support extended duration • Timing: protein C/S/antithrombin are unreliable during acute clot or on heparin/Warfarin
Q3. Patient will start heparin: calculate her doses.	• UFH 80 units/kg IV bolus, then 18 units/kg/h (actual body weight; caps per institutional protocol) • Adjust to aPTT or anti-Xa (0.3–0.7 units/mL)
Q4. Parameters to monitor after starting heparin?	• aPTT/anti-Xa: 6 h after start and after each change, then daily • CBC: Hgb, Hct, platelets baseline then every 2–3 days (HIT days 5–14) • Baseline: PT/INR, SCr, LFTs • Bleeding signs; K with prolonged use
Q5. Hypercoagulability positive; bridging heparin with Warfarin 10 mg. Target INR?	• INR 2.0–3.0 (target 2.5) • APS: Warfarin is preferred over DOACs
Q6. Day 3: INR 4.5, no bleeding. What to do with Warfarin and heparin?	• Warfarin: hold, then resume at a lower dose when INR is in range (10 mg starting dose was too high for her) • Heparin: an early high INR reflects factor VII loss, not full anticoagulation (factor II half-life 72 h); complete the **≥5-day overlap** and stop only when INR ≥2 for 24 h • Vitamin K: not routine without bleeding; oral 1–2.5 mg only if high bleeding risk • Check INR daily
Q7. She is discharged. When will you follow up?	• INR within 3–5 days after discharge (1–3 days if unstable) • Then weekly until stable

Question	Answer
Q8. After 2 months INR 2.4. What to do?	• In range: continue the same dose • Interval: extend gradually (every 4 weeks, up to 12 weeks if consistently stable)
Q9. Duration of Warfarin?	• First VTE with thrombophilia/unprovoked: at least 3 months, then reassess for extended therapy • Extend if not high bleeding risk (APS usually long term); reassess yearly
Q10. (2024 version) INR is 1.6 on Warfarin: what is the cause, and what do you do?	• **Azathioprine** can reduce the anticoagulant effect of Warfarin (interaction), so INR falls despite stable dosing • Check adherence, diet (Vitamin K) and other inducers; increase the weekly Warfarin dose by about **15%** (10–20%) and recheck INR in 3–5 days; keep target INR 2–3
Q11. Azathioprine is stopped: what do you do with Warfarin?	• The inducing effect fades over 1–2 weeks and INR will rise: **reduce Warfarin by about 15%** (back to the earlier dose), recheck INR in 3–5 days then weekly until stable • Antiphospholipid syndrome: stay on Warfarin (INR 2–3; **no DOACs**, especially if triple-positive or arterial events); pregnancy loss: LMWH + low-dose Aspirin in pregnancy
Also be ready for (exam add-ons)	• Goals: stop clot propagation, prevent death and recurrence, restore hemodynamics (if massive), prevent post-thrombotic syndrome and CTEPH • Non-pharm: early ambulation, compression stockings only for symptoms, hydration, stop smoking, weight control, review hormonal therapy (stop estrogen-containing contraceptives) • First-line: Enoxaparin 1 mg/kg q12h (or 1.5 mg/kg daily) or UFH 80 units/kg bolus then 18 units/kg/h (massive PE, planned fibrinolysis); DOAC: Apixaban 10 mg BID × 7 days then 5 mg BID, Rivaroxaban 15 mg BID × 21 days then 20 mg daily; massive PE: Alteplase 100 mg over 2 h; Warfarin INR 2–3 • Pregnancy: LMWH is the drug of choice (weight-based, no routine anti-Xa), UFH if delivery or fibrinolysis near; Warfarin and DOACs contraindicated; continue until 6 weeks postpartum (minimum 3 months total); Warfarin and Enoxaparin are safe during breastfeeding • Renal/dialysis: Enoxaparin 1 mg/kg q24h if CrCl <30 (anti-Xa monitoring); prefer UFH in dialysis; Apixaban needs no adjustment for VTE; Dabigatran, Edoxaban and Rivaroxaban avoided in severe CKD • Liver: avoid DOACs in Child-Pugh C; Warfarin has a high baseline INR; LMWH acceptable • HF/cancer: LMWH or Apixaban/Rivaroxaban/Edoxaban in cancer; avoid Warfarin if possible; Dabigatran with GI cancers carries more bleeding • Interactions: Carbamazepine, Phenytoin, Rifampin induce Warfarin and DOAC metabolism (use LMWH or an alternative anticonvulsant); estrogen-containing contraceptives are contraindicated after VTE (use IUD or progestin) • Counseling: injection technique and site rotation, bleeding precautions, avoid NSAIDs/Aspirin, signs of recurrence, duration (≥3 months; indefinite if unprovoked or recurrent), Warfarin diet and INR schedule • Thrombophilia work-up: antiphospholipid syndrome (lupus anticoagulant, anticardiolipin, β2-glycoprotein): avoid DOACs and use Warfarin (target 2–3) when triple-positive; send before heparin when possible

Case 7: Pregnancy with Sub-therapeutic Enoxaparin ★ Repeated case, also in 2018–2019

Case details: Pregnant woman with a history of DVT on Enoxaparin 60 mg BID. Today anti-Xa level is below range.

Question	Answer
Q1. Your recommendation regarding the dose, and counseling?	• Verify the sample: peak **4–6 h after the 3rd or 4th dose**; check adherence and technique • Targets: treatment q12h 0.6–1.0 units/mL; once daily 1.0–2.0; prophylaxis 0.2–0.5 • If truly low: increase dose ~10–25% (eg 60 → 70 mg q12h, or 1 mg/kg on current weight); repeat peak after the 3rd–4th new dose • Why low: weight gain, larger volume of distribution, higher renal clearance in pregnancy • Monitor: CBC (platelets), SCr • Counsel: SC technique, rotate sites, do not expel the air bubble or rub; bleeding signs; safe in breastfeeding • Delivery plan: stop 24 h before planned delivery/neuraxial anesthesia (or switch to UFH near term) • Avoid Warfarin and DOACs in pregnancy
Also be ready for (exam add-ons)	• Goals: prevent or treat VTE in pregnancy with the fewest risks to mother and fetus; keep anti-Xa in range • Non-pharm: mobilization, hydration, compression, avoid prolonged immobility, plan delivery and neuraxial anesthesia timing • First-line: Enoxaparin 1 mg/kg q12h (treatment) with peak anti-Xa 0.6–1.0 IU/mL drawn 4–6 h after the 3rd or 4th dose; prophylaxis 40 mg daily with peak 0.2–0.6 IU/mL; weight changes may need dose increases (level 0.25 = sub-therapeutic: increase dose and repeat the level) • Pregnancy: LMWH is safe; Warfarin (embryopathy weeks 6–12) and DOACs contraindicated; switch to UFH or stop LMWH 24 h before planned delivery; restart 4–6 h after vaginal or 6–12 h after cesarean; continue ≥6 weeks postpartum • Renal/dialysis: accumulation if CrCl <30 (use UFH or dose-reduce with anti-Xa monitoring) • Liver: platelets and LFTs (HELLP); LMWH acceptable • Counseling: self-injection technique, rotate sites, bruising is common, bleeding and labor signs (call early), regional anesthesia timing, avoid NSAIDs, breastfeeding is safe; repeat anti-Xa before the next appointment (about 1–2 weeks after the dose change)

Case 8: Atrial Fibrillation then STEMI ★ Repeated case, also in 2024, 2021, 2018–2019

Case details: 69-year-old with AF, HTN, DM on Diltiazem 180 mg, Flecainide, Metformin 1 g BID, Warfarin 6 mg (INR 2.7). HR 90–95 (goal <110).

Question	Answer
Q1. Risk factors for stroke?	• Age 65–74 • HTN • DM • AF • Vascular disease (after MI)
Q2. Calculate CHA2DS2-VASc and HAS-BLED.	• CHA2DS2-VASc: HTN 1 + DM 1 + Age 65–74 1 = 3; **4** after MI • HAS-BLED: Elderly 1 = 1; 2 once an antiplatelet is added
Q3. Symptoms (tachycardia, chest discomfort): what will you do? New target HR?	• Exclude ischemia: ECG, troponin • Tighten rate control: target **<80 at rest** when symptomatic • Switch Diltiazem to beta-blocker if CAD suspected; reconsider Flecainide (CAD)
Q4. Develops STEMI treated with PCI: what medication changes?	• Stop Flecainide (class Ic contraindicated post-MI) • Diltiazem → beta-blocker (Metoprolol succinate or Bisoprolol) • Warfarin → Apixaban + Clopidogrel; Aspirin ≤1 week • Add high-intensity statin; hold Metformin 48 h after contrast • Rhythm options: Amiodarone, Dofetilide, Sotalol • Doses: ◦ Apixaban 5 mg BID (2.5 mg if ≥2 of age ≥80, weight ≤60 kg, SCr ≥1.5) ◦ Clopidogrel 75 mg daily for 12 months, then OAC alone ◦ Aspirin 81 mg only until discharge or ≤1 week (max 30 days) ◦ Metoprolol succinate 25–50 mg daily, titrate to HR ◦ Atorvastatin 80 mg daily • Monitor: bleeding and Hgb, SCr (DOAC dosing) at least yearly, HR/BP, ECG (QTc if antiarrhythmic), lipids 4–8 weeks
Q5. Long-term medications post-MI?	• OAC + Clopidogrel to 12 months, then OAC alone • Beta-blocker • High-intensity statin ± Ezetimibe • ACE inhibitor/ARB (HTN, DM) • SL Nitroglycerin PRN • Cardiac rehabilitation • Doses: ◦ Apixaban 5 mg BID (2.5 mg if ≥2 of age ≥80, weight ≤60 kg, SCr ≥1.5) ◦ Clopidogrel 75 mg daily for 12 months, then OAC alone ◦ Atorvastatin 80 mg daily ◦ Metoprolol succinate 25–200 mg daily • Monitor: bleeding and Hgb, SCr (DOAC dosing) at least yearly, HR/BP, ECG (QTc if antiarrhythmic), lipids 4–8 weeks
Q6. Warfarin changed to Apixaban: how to convert?	• Stop Warfarin; start Apixaban when INR <2.0
Q7. Recommended Apixaban dose?	• Apixaban 5 mg BID
Q8. When to adjust the Apixaban dose? Adjusted dose? Applicable here?	• 2.5 mg BID if ≥2 of: age ≥80, weight ≤60 kg, SCr ≥1.5 mg/dL • Also: strong dual CYP3A4/P-gp inhibitors • Not applicable to this patient (69 years)
Also be ready for (exam add-ons)	• Goals: prevent stroke (anticoagulation), control rate or rhythm, treat the cause, improve symptoms and prevent HF/tachycardiomyopathy • Non-pharm: weight loss, alcohol reduction, treat sleep apnea, exercise, BP control, smoking cessation, avoid stimulants (caffeine, decongestants); catheter ablation if drug failure • First-line: anticoagulant by CHA2DS2-VASc (men ≥2, women ≥3): Apixaban 5 mg BID (2.5 mg if 2 of age ≥80, weight ≤60 kg, SCr ≥1.5), Rivaroxaban 20 mg with the evening meal (15 mg if CrCl 15–50), or Warfarin INR 2–3 (valvular AF); rate control: Metoprolol/Bisoprolol or Diltiazem (not in HFrEF); rhythm: Flecainide/Propafenone (no structural disease), Amiodarone/Dofetilide (HF, CAD) • Pregnancy: DOACs and Warfarin contraindicated; use LMWH (Enoxaparin 1 mg/kg q12h) or UFH; rate control with Metoprolol or Labetalol; cardioversion is safe; avoid Amiodarone, Dronedarone • Renal/dialysis: Apixaban 5 mg BID (2.5 mg if age ≥80 or weight ≤60) in HD; Dabigatran avoid if CrCl <30; Rivaroxaban 15 mg if CrCl 15–50, avoid <15; Edoxaban 30 mg if CrCl 15–50; Dofetilide/Sotalol adjust to CrCl • Liver: avoid DOACs in Child-Pugh C (Rivaroxaban avoided in B and C); Amiodarone monitor LFTs; Warfarin baseline INR is high • HF: avoid Flecainide, Propafenone, Dronedarone, Sotalol and Diltiazem/Verapamil in HFrEF; use Amiodarone or Dofetilide; β-blocker for rate • GI (PUD/GERD): PPI when combined with antiplatelets; bleeding risk; HAS-BLED corrections (stop NSAIDs, control BP) • Counseling: adherence (DOAC half-life is short), bleeding signs, Rivaroxaban with food, Warfarin: steady Vitamin K intake and INR checks, drug interactions (Amiodarone, Clarithromycin, Rifampin,

Question	Answer
	Carbamazepine), missed dose rules, show the card to all clinicians • Side scenario (post-MI): stop Flecainide/Propafenone (CAST) and Diltiazem if EF reduced; use a β-blocker, Amiodarone or Dofetilide; with PCI use Apixaban + Clopidogrel (stop Aspirin after 1 week to 1 month), then anticoagulant alone at 12 months; high-intensity statin

Case 9: Dyslipidemia, DM and Anticoagulation (Mitral Valve Replacement)

Case details: Obese patient with mitral valve replacement 6 years ago; HTN, DM, DLP on Metformin 1 g BID, Lisinopril 20 mg, Warfarin (wants to stop frequent monitoring). A1C 8.5%, FBG 110, BP 146/80, ASCVD risk 14%.

Question	Answer
Q1. Recommendation regarding DLP treatment?	• DM aged 40–75 with multiple risk factors (HTN, obesity) → high-intensity statin ◦ Atorvastatin 40–80 mg or Rosuvastatin 20–40 mg daily • Warfarin interaction: check INR after starting (Rosuvastatin can raise INR) • Monitor: lipid panel 4–12 weeks, INR within 1 week of starting, LFT baseline, muscle symptoms
Q2. Side effects of statins; how to investigate and manage?	• Side effects: muscle symptoms (myalgia → myopathy → rhabdomyolysis), raised transaminases, small rise in new-onset DM, rare immune-mediated necrotizing myopathy • Investigate: symptom pattern (symmetric, proximal), CK, TSH, renal function, interacting drugs • Mild–moderate: hold until resolved, rechallenge same or lower dose, a different statin, or alternate-day dosing • Severe or CK >10× ULN: stop; manage rhabdomyolysis • ALT >3× ULN persistent: stop and evaluate
Q3. Target LDL?	• LDL-C reduction ≥50% • ADA: LDL-C <70 mg/dL in DM aged 40–75 with ≥1 ASCVD risk factor
Q4. Target A1C? What anti-diabetic class will you add, with examples, and why?	• Target A1C <7% (individualized) • Add GLP-1 RA: Semaglutide 0.25 mg SC weekly × 4 weeks → 0.5 → 1 → 2 mg; Dulaglutide; Liraglutide • or GIP/GLP-1 RA: Tirzepatide 2.5 mg weekly, increase by 2.5 mg every 4 weeks • Why: obesity (weight loss), CV benefit, low hypoglycemia risk, A1C lowering • Alternative: SGLT2i Empagliflozin 10 mg for CV/renal protection • Monitor: A1C every 3 months until at goal, GI tolerance, weight, signs of pancreatitis; screen for personal/family history of MTC or MEN2 before GLP-1 RA • Non-pharm: medical nutrition therapy, ≥150 min/week activity, weight loss ≥5–10%
Q5. Based on ADA, what is the target BP? Why?	• <130/80 mm Hg if safely achievable • Why: lowers CV events and microvascular complications in DM • Now 146/80: increase Lisinopril or add Amlodipine or a thiazide
Q6. Can we change Warfarin to an anticoagulant with less monitoring? Why?	• Depends on the valve type • Mechanical valve: No. **DOACs are contraindicated** (RE-ALIGN: more thrombosis and bleeding with Dabigatran) ◦ Keep Warfarin, INR 3.0 (2.5–3.5) for mitral; consider INR home self-testing to reduce visits • Bioprosthetic valve >3–6 months: a DOAC is acceptable if AF is the indication
Also be ready for (exam add-ons)	• Goals: protect the mechanical valve from thrombosis (INR 2.5–3.5 mitral), control glucose (A1C <7%), BP <130/80 and LDL <70 • Non-pharm: diet and weight loss, exercise, steady Vitamin K intake, smoking and alcohol limits, foot/eye/kidney screening, INR clinic with a home INR meter for the patient who dislikes visits • First-line: Warfarin only (DOACs are contraindicated with mechanical valves); DM: add SGLT2 inhibitor or GLP-1 RA with CV benefit; high-intensity statin; ACE inhibitor/ARB for BP • Pregnancy: mechanical valve in pregnancy is very high risk: LMWH twice daily with anti-Xa monitoring or Warfarin ≤5 mg/day (weeks 6–12 risk) decided by a specialist team; avoid DOACs, statins, ACE inhibitors/ARBs, SGLT2 inhibitors • Renal/dialysis: Metformin stop if eGFR <30; SGLT2 inhibitor for kidney protection; Warfarin more sensitive in CKD • Liver: higher INR variability; Pioglitazone and statin caution • Counseling: INR goal and schedule, home INR testing, dental/endocarditis prophylaxis (Amoxicillin 2 g before procedures), bleeding signs, avoid NSAIDs, report new drugs, never stop Warfarin

2021

Case 1: Atrial Fibrillation then STEMI ★ Repeated case, also in 2024, 2022–2023, 2018–2019

Case details: 65-year-old with DM, HTN, AF on Flecainide, Diltiazem, Lisinopril, Metformin. Later developed STEMI.

Question	Answer
Q1. Calculate CHA2DS2-VASc and HAS-BLED.	• CHA2DS2-VASc: Age 65 1 + HTN 1 + DM 1 = 3; 4 after MI • HAS-BLED: Elderly 1; +1 for antiplatelet after PCI = 2 • Note: he is not anticoagulated → start OAC

Question	Answer
Q2. He developed STEMI: what changes to AF management?	• Stop Flecainide (class Ic in CAD) • Diltiazem → beta-blocker • Start DOAC (Apixaban) + Clopidogrel; Aspirin ≤1 week; then OAC alone after 12 months • Rhythm options: Amiodarone, Dofetilide, Sotalol • Add high-intensity statin; hold Metformin 48 h after contrast • Doses: ◦ Apixaban 5 mg BID (2.5 mg if ≥2 of age ≥80, weight ≤60 kg, SCr ≥1.5) ◦ Clopidogrel 75 mg daily for 12 months, then OAC alone ◦ Aspirin 81 mg only until discharge or ≤1 week (max 30 days) ◦ Metoprolol succinate 25–50 mg daily, titrate to HR ◦ Atorvastatin 80 mg daily • Monitor: bleeding and Hgb, SCr (DOAC dosing) at least yearly, HR/BP, ECG (QTc if antiarrhythmic), lipids 4–8 weeks
Also be ready for (exam add-ons)	• Goals: prevent stroke (anticoagulation), control rate or rhythm, treat the cause, improve symptoms and prevent HF/tachycardiomyopathy • Non-pharm: weight loss, alcohol reduction, treat sleep apnea, exercise, BP control, smoking cessation, avoid stimulants (caffeine, decongestants); catheter ablation if drug failure • First-line: anticoagulant by CHA2DS2-VASc (men ≥2, women ≥3): Apixaban 5 mg BID (2.5 mg if 2 of age ≥80, weight ≤60 kg, SCr ≥1.5), Rivaroxaban 20 mg with the evening meal (15 mg if CrCl 15–50), or Warfarin INR 2–3 (valvular AF); rate control: Metoprolol/Bisoprolol or Diltiazem (not in HFrEF); rhythm: Flecainide/Propafenone (no structural disease), Amiodarone/Dofetilide (HF, CAD) • Pregnancy: DOACs and Warfarin contraindicated; use LMWH (Enoxaparin 1 mg/kg q12h) or UFH; rate control with Metoprolol or Labetalol; cardioversion is safe; avoid Amiodarone, Dronedarone • Renal/dialysis: Apixaban 5 mg BID (2.5 mg if age ≥80 or weight ≤60) in HD; Dabigatran avoid if CrCl <30; Rivaroxaban 15 mg if CrCl 15–50, avoid <15; Edoxaban 30 mg if CrCl 15–50; Dofetilide/Sotalol adjust to CrCl • Liver: avoid DOACs in Child-Pugh C (Rivaroxaban avoided in B and C); Amiodarone monitor LFTs; Warfarin baseline INR is high • HF: avoid Flecainide, Propafenone, Dronedarone, Sotalol and Diltiazem/Verapamil in HFrEF; use Amiodarone or Dofetilide; β-blocker for rate • GI (PUD/GERD): PPI when combined with antiplatelets; bleeding risk; HAS-BLED corrections (stop NSAIDs, control BP) • Counseling: adherence (DOAC half-life is short), bleeding signs, Rivaroxaban with food, Warfarin: steady Vitamin K intake and INR checks, drug interactions (Amiodarone, Clarithromycin, Rifampin, Carbamazepine), missed dose rules, show the card to all clinicians • Side scenario (post-MI): stop Flecainide/Propafenone (CAST) and Diltiazem if EF reduced; use a β-blocker, Amiodarone or Dofetilide; with PCI use Apixaban + Clopidogrel (stop Aspirin after 1 week to 1 month), then anticoagulant alone at 12 months; high-intensity statin

2020

Cases 1 and 3: HFrEF (Repeated Station) ★ Repeated case, also in 2024, 2020

> **Case details:** EF <30%, signs and symptoms with minimal work. On Amlodipine.

Question	Answer
Q1. Stage and NYHA class (number and letter)? Why?	• Stage C: structural heart disease (EF <30%) with symptoms • NYHA III: symptoms with less than ordinary (minimal) activity; comfortable at rest
Q2. What to add (all GDMT, anti-HF medications)?	• ARNI Sacubitril/valsartan (or ACE inhibitor/ARB) • Beta-blocker Bisoprolol, Carvedilol or Metoprolol succinate • MRA Spironolactone or Eplerenone • SGLT2i Dapagliflozin or Empagliflozin • Loop diuretic for congestion • Doses: ◦ Sacubitril/valsartan 24/26 mg BID (49/51 if on standard-dose ACE inhibitor/ARB) → 97/103 mg BID ◦ Bisoprolol 1.25 → 10 mg daily, Carvedilol 3.125 → 25 mg BID, or Metoprolol succinate 12.5–25 → 200 mg daily ◦ Spironolactone 12.5–25 mg daily (eGFR >30, K <5.0) ◦ Dapagliflozin or Empagliflozin 10 mg daily ◦ Furosemide 20–40 mg daily–BID, titrated to euvolemia • Non-pharm: sodium <2300 mg/day, daily weights, exercise/cardiac rehab, vaccines, avoid NSAIDs and non-DHP CCBs
Q3. Why?	• ARNI/ACE inhibitor/ARB: lower mortality, hospitalization, remodeling • Beta-blocker: lower mortality and sudden death • MRA: lower mortality and hospitalization • SGLT2i: lower CV death or worsening HF • Diuretic: symptom relief only

Question	Answer
Q4. Monitoring: safety and efficacy?	• Efficacy: symptoms/NYHA, weight, edema, BP, HR, NT-proBNP, EF • ARNI/ACE inhibitor/ARB: SCr, K at 1–2 weeks; hypotension, angioedema • Beta-blocker: HR, BP, fluid status, fatigue • MRA: K, SCr at ~1 week, 1 month, 6 months • SGLT2i: SCr, volume, genital infection, ketoacidosis • Diuretic: electrolytes, SCr, weight
Q5. What to target?	• Target or maximally tolerated GDMT doses • BP <130/80 • Euvolemia • Symptom improvement • LVEF improvement
Q6. What to modify: Amlodipine?	• Stop Amlodipine: no HF benefit, worsens peripheral edema, takes BP room from GDMT
Q7. Cut-off point of kidney function for Spironolactone?	• eGFR must be **>30 mL/min/1.73 m²** (and SCr ≤2.5 men, ≤2.0 women) • eGFR 30–49: lower starting dose (12.5 mg daily or every other day)
Q8. Cut-off point for potassium?	• Start only if K <5.0 mEq/L • K >5.5: halve the dose or stop
Q9. When is the next clinic visit?	• 1–2 weeks after each start or titration (labs, BP, HR, symptoms)
Q10. When to check ECHO?	• 3–6 months after GDMT reaches target/maximally tolerated doses • ICD decision after ≥3 months of optimized GDMT
Q11. What do you check in the ECHO?	• LVEF • LV dimensions/dilation • Wall thickness (LVH) • Wall motion • Valves • Diastolic function • RV function
Also be ready for (exam add-ons)	• Goals: reduce mortality and HF hospitalization, relieve congestion and symptoms, prevent remodeling, improve exercise tolerance and quality of life • Non-pharm: sodium <2–3 g/day, fluid 1.5–2 L only if hyponatremia or advanced HF, daily weights, exercise or cardiac rehabilitation, vaccines (influenza, pneumococcal, COVID-19), stop smoking, limit alcohol, avoid NSAIDs • First-line: 4 pillars: ARNI Sacubitril/valsartan 24/26 mg BID (or ACE inhibitor/ARB), Bisoprolol 1.25 mg (or Carvedilol 3.125 mg BID, Metoprolol succinate 12.5–25 mg) when euvolemic, Spironolactone 12.5–25 mg (eGFR >30, K <5), Dapagliflozin 10 mg or Empagliflozin 10 mg; Furosemide 20–40 mg only for congestion • Pregnancy: ACE inhibitor, ARB, ARNI, MRA and SGLT2 inhibitors are contraindicated; use Metoprolol, Labetalol, Hydralazine + Isosorbide dinitrate and a loop diuretic; peripartum cardiomyopathy: add Bromocriptine and anticoagulation per specialist; counsel contraception • Renal/dialysis: check SCr and K (ARNI, ACE inhibitor, MRA); avoid MRA if eGFR <30 or K >5; Empagliflozin/Dapagliflozin can start at eGFR ≥20; higher loop-diuretic doses needed; Digoxin reduce in CKD • Liver: avoid ARNI in Child-Pugh C; Carvedilol avoid in severe disease; congestive hepatopathy improves with decongestion • GI (PUD/GERD): if on antiplatelet or anticoagulant add a PPI (Pantoprazole preferred with Clopidogrel); avoid NSAIDs • Counseling: ARNI: dizziness, angioedema, hyperkalemia, 36-hour ACE-inhibitor washout; β-blocker: fatigue, do not stop abruptly; Spironolactone: avoid potassium salt substitutes, gynecomastia; SGLT2 inhibitor: genital infection, hydration, hold on sick days; weigh daily and call if gain >1 kg in a day or 2 kg in a week

Case 2: Heart Failure (Detailed) ★ Repeated case, also in 2024, 2020

Case details: Heart failure case.

Question	Answer
Q1. Classification of the patient's HF?	• ACC/AHA stage: A at risk; B structural disease without symptoms; C symptoms; D advanced • NYHA class I–IV by functional limitation • By EF: HFrEF ≤40%, HFmrEF 41–49%, HFpEF ≥50%, HFimpEF (previously ≤40%, now >40%)
Q2. Evidence of structural heart disease from the ECHO?	• Reduced EF • LV dilation • Regional wall motion abnormality • Valve disease • LVH • Raised filling pressures

Question	Answer
Q3. What medications will you start?	• ARNI (or ACE inhibitor/ARB) • Beta-blocker: only Bisoprolol, Carvedilol, Metoprolol succinate • MRA, SGLT2i • Loop diuretic for congestion • Doses: ◦ Sacubitril/valsartan 24/26 mg BID (49/51 if on standard-dose ACE inhibitor/ARB) → 97/103 mg BID ◦ Bisoprolol 1.25 → 10 mg daily, Carvedilol 3.125 → 25 mg BID, or Metoprolol succinate 12.5–25 → 200 mg daily ◦ Spironolactone 12.5–25 mg daily (eGFR >30, K <5.0) ◦ Dapagliflozin or Empagliflozin 10 mg daily ◦ Furosemide 20–40 mg daily–BID, titrated to euvolemia • Monitor: BP, HR, SCr and K 1–2 weeks after each start/titration; weight and congestion each visit • If EF >40% (HFmrEF/HFpEF): SGLT2 inhibitor first (Dapagliflozin or Empagliflozin 10 mg); consider Finerenone (FINEARTS-HF), an ARNI or MRA, and Semaglutide or Tirzepatide when obesity is present (STEP-HFpEF, SUMMIT)
Q4. What will you check before starting Spironolactone?	• eGFR >30 • K <5.0 • SCr ≤2.5 (men) / ≤2.0 (women) • Other K-raising drugs
Q5. Are you going to start with full doses?	• No: start low and titrate to target or maximally tolerated dose
Q6. What will you monitor for each class?	• ARNI/ACE inhibitor/ARB: BP, SCr, K, angioedema, cough • Beta-blocker: HR, BP, fluid retention, fatigue • MRA: K, SCr, gynecomastia • SGLT2i: SCr, volume, genital/urinary infection, euglycemic DKA • Loop: weight, UOP, electrolytes
Q7. Monitoring parameters (overall)?	• Symptoms/NYHA • Weight and edema • BP and HR • SCr, K, Na, Mg • NT-proBNP (BNP rises on ARNI) • Echo
Q8. For the aldosterone antagonist, with timing of monitoring?	• K and SCr: ~1 week, 1 month and 6 months after starting or a dose change, then periodically • K >5.5: halve or stop
Also be ready for (exam add-ons)	• Goals: reduce mortality and HF hospitalization, relieve congestion and symptoms, prevent remodeling, improve exercise tolerance and quality of life • Non-pharm: sodium <2–3 g/day, fluid 1.5–2 L only if hyponatremia or advanced HF, daily weights, exercise or cardiac rehabilitation, vaccines (influenza, pneumococcal, COVID-19), stop smoking, limit alcohol, avoid NSAIDs • First-line: 4 pillars: ARNI Sacubitril/valsartan 24/26 mg BID (or ACE inhibitor/ARB), Bisoprolol 1.25 mg (or Carvedilol 3.125 mg BID, Metoprolol succinate 12.5–25 mg) when euvolemic, Spironolactone 12.5–25 mg (eGFR >30, K <5), Dapagliflozin 10 mg or Empagliflozin 10 mg; Furosemide 20–40 mg only for congestion • Pregnancy: ACE inhibitor, ARB, ARNI, MRA and SGLT2 inhibitors are contraindicated; use Metoprolol, Labetalol, Hydralazine + Isosorbide dinitrate and a loop diuretic; peripartum cardiomyopathy: add Bromocriptine and anticoagulation per specialist; counsel contraception • Renal/dialysis: check SCr and K (ARNI, ACE inhibitor, MRA); avoid MRA if eGFR <30 or K >5; Empagliflozin/Dapagliflozin can start at eGFR ≥20; higher loop-diuretic doses needed; Digoxin reduce in CKD • Liver: avoid ARNI in Child-Pugh C; Carvedilol avoid in severe disease; congestive hepatopathy improves with decongestion • GI (PUD/GERD): if on antiplatelet or anticoagulant add a PPI (Pantoprazole preferred with Clopidogrel); avoid NSAIDs • Counseling: ARNI: dizziness, angioedema, hyperkalemia, 36-hour ACE-inhibitor washout; β-blocker: fatigue, do not stop abruptly; Spironolactone: avoid potassium salt substitutes, gynecomastia; SGLT2 inhibitor: genital infection, hydration, hold on sick days; weigh daily and call if gain >1 kg in a day or 2 kg in a week

Case 4: Stage 2 HTN and Prediabetic Smoker ★ Repeated case, also in 2024

Case details: Stage 2 HTN, prediabetes, smoker.

Question	Answer
Q1. What are the patient's risk factors?	• Smoking • Prediabetes • Obesity/overweight • Sedentary lifestyle • High-sodium diet

Question	Answer
Q2. Non-pharmacological interventions, and how much weight loss is needed?	• Weight loss ≥5% of body weight (~1 mm Hg per kg) • DASH diet • Sodium <2300 mg/day (ideally <1500) • Regular exercise • Limit alcohol • Smoking cessation • Pharm: stage 2 → two agents, eg Lisinopril 10 mg + Amlodipine 5 mg daily • Monitor: BP in 4 weeks, SCr and K 1–2 weeks, weight, A1C/FBG yearly
Q3. Nicotine replacement: dosage forms, patch dose, gum dose?	• Forms: patch, gum, lozenge, nasal spray, inhaler • Patch: >10 cig/day 21 mg × 6 weeks → 14 mg × 2 → 7 mg × 2; ≤10 cig/day 14 mg × 6 weeks → 7 mg × 2 • Gum: 4 mg if first cigarette within 30 min of waking, otherwise 2 mg; q1–2h weeks 1–6, q2–4h weeks 7–9, q4–6h weeks 10–12
Q4. Complications of HTN and DM?	• HTN: stroke, MI, HF, CKD, retinopathy • DM: retinopathy, nephropathy, neuropathy, ASCVD, foot ulcers, DKA/HHS, hypoglycemia
Q5. Which vaccines (none in 20 years)?	• Tdap then Td/Tdap every 10 years • Influenza yearly • Pneumococcal PCV20 or PCV21 (smoker) • COVID-19 • Hepatitis B (19–59 y) • Zoster ≥50 y
Also be ready for (exam add-ons)	• Goals: BP <130/80, prevent stroke, MI, HF and kidney disease, control risk factors • Non-pharm: DASH diet (−8–14 mm Hg), sodium <1.5–2.3 g/day (−2–8), weight loss (−1 mm Hg per kg), exercise 150 min/week (−4–9), alcohol limit, smoking cessation, stress management • First-line: stage 2: two agents (ACE inhibitor/ARB + CCB or thiazide) e.g., Lisinopril 10 mg + Amlodipine 5 mg or Hydrochlorothiazide 12.5–25 mg; DM or albuminuria: ACE inhibitor/ARB • Pregnancy: avoid ACE inhibitors, ARBs, direct renin inhibitors and thiazides in planning; use Labetalol 100–400 mg BID, Nifedipine ER 30–60 mg or Methyldopa; low-dose Aspirin from 12 weeks if preeclampsia risk • Renal/dialysis: ACE inhibitor/ARB for albuminuria (monitor SCr up to 30% rise and K); thiazides weaker at eGFR <30 (use loop); avoid K-sparing in advanced CKD • Liver: Labetalol hepatotoxicity monitoring; avoid non-selective β-blockers in decompensated cirrhosis (except Carvedilol for varices) • HF/DM: ARNI, β-blocker, MRA and SGLT2 inhibitor in HFrEF; avoid non-DHP CCB in HFrEF and Pioglitazone • Counseling: home BP monitoring, adherence, ACE inhibitor cough/angioedema, Amlodipine edema, thiazide gout/hypokalemia, rise slowly, avoid NSAIDs/decongestants/licorice • Smoking cessation: counsel (5 A's); Varenicline 0.5 mg daily × 3 days, 0.5 mg BID × 4 days, then 1 mg BID × 12 weeks, or combination NRT (patch 21/14/7 mg + gum/lozenge 2–4 mg); Bupropion SR 150 mg daily × 3 days then 150 mg BID, also first-line (Varenicline is the most effective single agent); monitor BP on NRT and Bupropion • Vaccines: influenza yearly, pneumococcal (PCV20/21), Td/Tdap every 10 years, COVID-19, hepatitis B, zoster

2018–2019

Case 1: Dyslipidemia (Oral) ★ Repeated case, also in 2024

Case details: Oral dyslipidemia station.

Question	Answer
Q1. Diet (type of diet)?	• Heart-healthy: DASH or Mediterranean • More: fruits, vegetables, whole grains, low-fat dairy • Less: sweets, sugary drinks, red meat • Fat: replace saturated with unsaturated (especially polyunsaturated)
Q2. Exercise (minutes/week)? Statin choice by intensity?	• Exercise: moderate–vigorous aerobic 3–4 times/week, ~40 min/session • High intensity: Atorvastatin 40–80 mg; Rosuvastatin 20–40 mg • Moderate: Atorvastatin 10–20; Rosuvastatin 5–10; Simvastatin 20–40; Pravastatin 40–80; Lovastatin 40–80; Fluvastatin XL 80 or 40 BID; Pitavastatin 1–4 • Low: Simvastatin 10; Pravastatin 10–20; Lovastatin 20; Fluvastatin 20–40 • DM aged <75 with ASCVD risk 14%: moderate intensity; high intensity if multiple risk factors

Question	Answer
Q3. PCSK9 inhibitor names and use?	• Mechanism: monoclonal antibodies bind PCSK9 → more LDL receptors recycled → LDL-C ↓ ~50–60% • Evolocumab: 140 mg SC q2 weeks or 420 mg monthly • Alirocumab: 75 mg SC q2 weeks, up to 150 mg q2 weeks (or 300 mg q4 weeks) • Inclisiran (siRNA): 284 mg SC at 0 and 3 months, then every 6 months • Side effects: injection-site reactions, nasopharyngitis, flu-like symptoms • Use: with statin ± Ezetimibe in ASCVD or FH not at goal • Newer oral option: Bempedoic acid 180 mg daily for statin intolerance or as add-on to Ezetimibe (CLEAR Outcomes)
Q4. Add PCSK9 or not? Aspirin for prevention?	• PCSK9: only for clinical ASCVD or FH above goal on max statin + Ezetimibe • Aspirin: yes for secondary prevention; not routine for primary prevention • BP goal: <130/80
Also be ready for (exam add-ons)	• Goals: lower LDL-C and ASCVD risk (primary: ≥30–49% reduction by risk; secondary: ≥50% and <70 mg/dL), reduce events and mortality; prevent pancreatitis if TG very high • Non-pharm: Mediterranean diet (saturated fat <7%, no trans fat), fiber 10–25 g, plant sterols, exercise 150 min/week, weight loss, smoking cessation, limit alcohol and sugar • First-line: risk 7.5–<20%: moderate-intensity statin (Atorvastatin 10–20 mg, Rosuvastatin 5–10 mg); ≥20% or ASCVD: high-intensity (Atorvastatin 40–80 mg, Rosuvastatin 20–40 mg); add Ezetimibe 10 mg then a PCSK9 inhibitor (Evolocumab 140 mg q2 weeks, Alirocumab 75–150 mg q2 weeks) if LDL above goal; TG >500: Fenofibrate • Pregnancy: statins and PCSK9 inhibitors are stopped before conception or as soon as pregnancy is confirmed (and during breastfeeding); Ezetimibe is avoided (limited data); Colesevelam is acceptable • Renal/dialysis: Atorvastatin and Fluvastatin need no adjustment; Rosuvastatin max 10 mg if CrCl <30; Simvastatin/Pravastatin start low; avoid Gemfibrozil with statins; Fenofibrate reduce • Liver: avoid in active liver disease; ALT >3× ULN: hold; do not use in decompensated cirrhosis • Interactions: Amiodarone → Simvastatin ≤20 mg; Diltiazem/Verapamil → Simvastatin ≤10 mg; Clarithromycin, azoles, Cyclosporine; Gemfibrozil + statin = rhabdomyolysis • Counseling: evening dosing for short half-life statins (Simvastatin, Lovastatin, Pravastatin), muscle pain with dark urine = stop and call, avoid large amounts of grapefruit juice (Simvastatin, Atorvastatin), adherence, diet, lipid panel 4–12 weeks then every 3–12 months

Case 2: Warfarin and DOAC Conversion

Case details: Conversions between Warfarin and DOACs; Apixaban dosing.

Question	Answer
Q1. Converting from Warfarin to a DOAC?	• Apixaban: start when INR <2.0 • Dabigatran: start when INR <2.0 • Rivaroxaban: start when INR <3.0 • Edoxaban: start when INR ≤2.5 • UFH infusion / LMWH / fondaparinux: start when INR <2.0
Q2. Converting Apixaban or Rivaroxaban to Warfarin?	• Stop the DOAC; start a parenteral anticoagulant + Warfarin when the next DOAC dose was due • Stop the parenteral agent when INR >2.0 • Edoxaban oral option: halve the dose, start Warfarin together; stop Edoxaban when INR ≥2.0 (INR drawn before the Edoxaban dose)
Q3. Converting Dabigatran to Warfarin (by CrCl)?	• CrCl ≥50: start Warfarin 3 days before stopping Dabigatran • CrCl 30–50: 2 days before • CrCl 15–30: 1 day before • CrCl <15: no recommendation • INR reflects Warfarin only after Dabigatran has been stopped ≥2 days
Q4. Apixaban dosing (by indication, age, weight, renal function)?	• First ask: indication; valvular (mechanical valve or moderate–severe mitral stenosis) → Warfarin, otherwise DOAC • AF: 5 mg BID; 2.5 mg BID if ≥2 of age ≥80, weight ≤60 kg, SCr ≥1.5 mg/dL • AF after PCI: same dose + Clopidogrel; Aspirin ≤1 week • VTE treatment: 10 mg BID × 7 days, then 5 mg BID; extended phase 2.5 mg BID after 6 months • Dialysis (AF): 5 mg BID; 2.5 mg BID if ≥80 y or ≤60 kg • Monitor: CBC, SCr, LFTs at baseline and at least yearly
Also be ready for (exam add-ons)	• Goals: change anticoagulant safely without a gap or overlap • Non-pharm: INR check before the switch, assess adherence, renal function, weight, age and interacting drugs • First-line: Warfarin → Apixaban/Dabigatran when INR <2; Rivaroxaban when INR <3; Edoxaban when INR ≤2.5; DOAC → Warfarin: overlap with a parenteral anticoagulant until INR ≥2 • Pregnancy: DOACs and Warfarin are contraindicated; use LMWH • Renal/dialysis: Apixaban 2.5 mg BID if 2 of age ≥80, weight ≤60 kg, SCr ≥1.5; Dabigatran avoid CrCl <30; Rivaroxaban avoid <15 • Liver: avoid DOACs in Child-Pugh C

Question	Answer
	• Counseling: no INR monitoring on DOACs but yearly CBC/SCr/LFTs; adherence; missed dose; interactions (Rifampin, Carbamazepine, azoles)

Case 3: Atrial Fibrillation (Direct) ★ Repeated case, also in 2024, 2022–2023, 2021

Case details: AF patient with HTN, DM, age 69.

Question	Answer
Q1. Risk factors for stroke in this patient?	• HTN • DM • Age 69 • AF
Q2. CHADS score?	• CHADS2: HTN 1 + DM 1 = 2 (age counts only at ≥75) • CHA2DS2-VASc: HTN 1 + DM 1 + Age 65–74 1 = 3
Q3. HAS-BLED score?	• Elderly >65: 1 (others depend on case data)
Q4. Patient develops STEMI: what medication changes?	• Stop class Ic (Flecainide); non-DHP CCB → beta-blocker • OAC + Clopidogrel, Aspirin ≤1 week • High-intensity statin • Doses: ◦ Apixaban 5 mg BID (2.5 mg if ≥2 of age ≥80, weight ≤60 kg, SCr ≥1.5) ◦ Clopidogrel 75 mg daily for 12 months, then OAC alone ◦ Aspirin 81 mg only until discharge or ≤1 week (max 30 days) ◦ Metoprolol succinate 25–50 mg daily, titrate to HR ◦ Atorvastatin 80 mg daily • Monitor: bleeding and Hgb, SCr (DOAC dosing) at least yearly, HR/BP, ECG (QTc if antiarrhythmic), lipids 4–8 weeks
Q5. Shifted from Warfarin to Apixaban: criteria for dose reduction?	• 2.5 mg BID if ≥2 of: age ≥80, weight ≤60 kg, SCr ≥1.5 mg/dL
Q6. How will you switch from Warfarin to Apixaban?	• Stop Warfarin; start Apixaban when INR <2.0
Also be ready for (exam add-ons)	• Goals: prevent stroke (anticoagulation), control rate or rhythm, treat the cause, improve symptoms and prevent HF/tachycardiomyopathy • Non-pharm: weight loss, alcohol reduction, treat sleep apnea, exercise, BP control, smoking cessation, avoid stimulants (caffeine, decongestants); catheter ablation if drug failure • First-line: anticoagulant by CHA2DS2-VASc (men ≥2, women ≥3): Apixaban 5 mg BID (2.5 mg if 2 of age ≥80, weight ≤60 kg, SCr ≥1.5), Rivaroxaban 20 mg with the evening meal (15 mg if CrCl 15–50), or Warfarin INR 2–3 (valvular AF); rate control: Metoprolol/Bisoprolol or Diltiazem (not in HFrEF); rhythm: Flecainide/Propafenone (no structural disease), Amiodarone/Dofetilide (HF, CAD) • Pregnancy: DOACs and Warfarin contraindicated; use LMWH (Enoxaparin 1 mg/kg q12h) or UFH; rate control with Metoprolol or Labetalol; cardioversion is safe; avoid Amiodarone, Dronedarone • Renal/dialysis: Apixaban 5 mg BID (2.5 mg if age ≥80 or weight ≤60) in HD; Dabigatran avoid if CrCl <30; Rivaroxaban 15 mg if CrCl 15–50, avoid <15; Edoxaban 30 mg if CrCl 15–50; Dofetilide/Sotalol adjust to CrCl • Liver: avoid DOACs in Child-Pugh C (Rivaroxaban avoided in B and C); Amiodarone monitor LFTs; Warfarin baseline INR is high • HF: avoid Flecainide, Propafenone, Dronedarone, Sotalol and Diltiazem/Verapamil in HFrEF; use Amiodarone or Dofetilide; β-blocker for rate • GI (PUD/GERD): PPI when combined with antiplatelets; bleeding risk; HAS-BLED corrections (stop NSAIDs, control BP) • Counseling: adherence (DOAC half-life is short), bleeding signs, Rivaroxaban with food, Warfarin: steady Vitamin K intake and INR checks, drug interactions (Amiodarone, Clarithromycin, Rifampin, Carbamazepine), missed dose rules, show the card to all clinicians • Side scenario (post-MI): stop Flecainide/Propafenone (CAST) and Diltiazem if EF reduced; use a β-blocker, Amiodarone or Dofetilide; with PCI use Apixaban + Clopidogrel (stop Aspirin after 1 week to 1 month), then anticoagulant alone at 12 months; high-intensity statin

Case 4: Pregnancy on Enoxaparin (Sub-therapeutic Level) ★ Repeated case, also in 2022–2023

Case details: Pregnant woman on Enoxaparin; anti-Xa level 0.25 units/mL.

Question	Answer
Q1. Recommendation about TDM?	• Target (treatment q12h): peak 0.6–1.0 units/mL → 0.25 is low • Confirm timing (peak 4–6 h after the 3rd–4th dose) and adherence • Increase dose ~10–25% (or 1 mg/kg q12h on current weight)

Question	Answer
Q2. When to repeat the level? Patient education about Enoxaparin?	• Repeat peak 4–6 h after the 3rd–4th dose of the new regimen • Technique: SC abdomen, rotate sites, do not expel the bubble or rub • Report bleeding or bruising • Before delivery: stop 24 h before planned induction/neuraxial anesthesia • Breastfeeding: safe
Also be ready for (exam add-ons)	• Goals: prevent or treat VTE in pregnancy with the fewest risks to mother and fetus; keep anti-Xa in range • Non-pharm: mobilization, hydration, compression, avoid prolonged immobility, plan delivery and neuraxial anesthesia timing • First-line: Enoxaparin 1 mg/kg q12h (treatment) with peak anti-Xa 0.6–1.0 IU/mL drawn 4–6 h after the 3rd or 4th dose; prophylaxis 40 mg daily with peak 0.2–0.6 IU/mL; weight changes may need dose increases (level 0.25 = sub-therapeutic: increase dose and repeat the level) • Pregnancy: LMWH is safe; Warfarin (embryopathy weeks 6–12) and DOACs contraindicated; switch to UFH or stop LMWH 24 h before planned delivery; restart 4–6 h after vaginal or 6–12 h after cesarean; continue ≥6 weeks postpartum • Renal/dialysis: accumulation if CrCl <30 (use UFH or dose-reduce with anti-Xa monitoring) • Liver: platelets and LFTs (HELLP); LMWH acceptable • Counseling: self-injection technique, rotate sites, bruising is common, bleeding and labor signs (call early), regional anesthesia timing, avoid NSAIDs, breastfeeding is safe; repeat anti-Xa before the next appointment (about 1–2 weeks after the dose change)

Case 5: Anticoagulation Clinic (Warfarin with Nasal Bleeding) ★ Repeated case, also in 2024

Case details: Warfarin started; at the next visit the patient has nasal bleeding.

Question	Answer
Q1. How to proceed? Switch the anticoagulant?	• Assess: severity, INR, Hgb • Minor epistaxis: local measures, hold dose; oral Vitamin K 2.5–5 mg if INR ≥4.5 • Major: Vitamin K 5–10 mg IV + 4F-PCC • Find the cause (interactions, diet, adherence) • Switch to DOAC if no mechanical valve/moderate–severe mitral stenosis and INR is labile (TTR <65–70%)
Also be ready for (exam add-ons)	• Goals: keep INR in range (2–3 for most indications; 2.5–3.5 mechanical mitral valve), stop bleeding, avoid thrombosis • Non-pharm: local pressure (10–15 min) for nasal bleeding, identify the cause (dose, interacting drug, infection, diet, alcohol), consistent Vitamin K intake, written dose calendar • First-line: INR >4.5–10 without bleeding: hold 1–2 doses, reduce weekly dose 10–20%; no routine Vitamin K unless high bleeding risk (Vitamin K 1–2.5 mg PO); major bleeding: Vitamin K 5–10 mg IV + 4-factor PCC • Pregnancy: Warfarin is teratogenic (6–12 weeks) and crosses the placenta: switch to LMWH before week 6 (mechanical valves: specialist decision); Warfarin is safe during breastfeeding • Renal/dialysis: higher bleeding risk, lower dose requirement; consider Apixaban in AF with dialysis (non-valvular) • Liver: baseline INR high, lower doses, closer monitoring • HF: congestion raises INR sensitivity; interacting drugs: Amiodarone (30–50% dose cut), Fluconazole, Metronidazole, TMP-SMX, quinolones, Rifampin (increase dose), Carbamazepine • Counseling: bleeding signs, steady greens, avoid alcohol binges and NSAIDs/Aspirin, report new drugs/antibiotics, never double a missed dose, INR 3–5 days after any change, carry a card

Case 6: ACS ★ Repeated case, also in 2025, 2024, 2022–2023

Case details: ACS case; patient on Metformin.

Question	Answer
Q1. What will you give in the ER?	• Aspirin 162–325 mg chewed • P2Y12 load: Ticagrelor 180 mg / Prasugrel 60 mg / Clopidogrel 600 mg • Anticoagulant • Nitroglycerin SL • Oxygen if SaO2 <90% • Oral beta-blocker within 24 h • High-intensity statin • Monitor: ECG, BP, HR, SaO2, pain, troponin, Hgb, SCr
Q2. Pain relief?	• First: Nitroglycerin 0.4 mg SL q5min × 3, then IV 10 mcg/min • If refractory: Morphine 2–4 mg IV q5–15 min (delays oral P2Y12 absorption) • Avoid NSAIDs (other than Aspirin) • Monitor: pain score, BP (hold NTG if SBP <90), respiratory rate and sedation with Morphine

Question	Answer
Q3. What will you give in the cath lab, with doses?	• UFH: 70–100 units/kg IV (50–70 with GP IIb/IIIa) • or Bivalirudin: 0.75 mg/kg then 1.75 mg/kg/h • GP IIb/IIIa (bailout): Eptifibatide 180 mcg/kg × 2 + 2 mcg/kg/min; Tirofiban 25 mcg/kg + 0.15 mcg/kg/min
Q4. On discharge: which beta-blocker and dose proven to improve survival, and for how long?	• Metoprolol: succinate 25 → 200 mg daily (tartrate 25–100 mg BID) • Carvedilol: 6.25 → 25 mg BID • Bisoprolol if HFrEF • Duration: long term if EF ≤40%, angina, arrhythmia or HTN; otherwise reassess after 1 year
Q5. Statin side effects, and when to stop?	• Side effects: muscle symptoms, raised transaminases, new-onset DM • Stop: CK >10× ULN with symptoms or rhabdomyolysis; persistent ALT >3× ULN
Q6. How to reduce the risk of contrast-induced nephropathy?	• **Hold Metformin ≥48 h after contrast** • IV isotonic saline • Low/iso-osmolar contrast, minimal volume • Stop nephrotoxins, avoid diuretics • Oral acetylcysteine optional
Also be ready for (exam add-ons)	• Goals: restore infarct-artery patency quickly (door-to-balloon ≤90 min, door-to-needle ≤30 min), limit infarct size, relieve ischemia, prevent arrhythmia, heart failure and reinfarction • Non-pharm: primary PCI (preferred), cardiac rehab, smoking cessation, Mediterranean diet, exercise 150 min/week, weight control, sodium restriction, vaccines (influenza), driving advice • First-line: Aspirin 162–325 mg chewed, P2Y12 inhibitor (Ticagrelor 180 mg or Prasugrel 60 mg or Clopidogrel 600 mg), UFH 70–100 units/kg in the cath lab, Atorvastatin 80 mg, oral β-blocker within 24 h if LVEF reduced, HF or ventricular arrhythmia and no shock (2025 ACC/AHA; reassess after 1 year if LVEF is preserved), Nitroglycerin SL for pain • Pregnancy: MI in pregnancy is rare: PCI preferred, avoid fibrinolysis if possible; low-dose Aspirin and Clopidogrel acceptable; avoid ACE inhibitors/ARBs/MRA and statins; Metoprolol and Labetalol are safe; UFH or Enoxaparin acceptable • Renal/dialysis: contrast nephropathy prevention (hydration, minimal contrast, hold Metformin and nephrotoxins); Enoxaparin 1 mg/kg q24h if CrCl <30; Bivalirudin: reduce the infusion if CrCl <30; Ticagrelor needs no adjustment • Liver: avoid Ticagrelor in severe hepatic impairment; statin caution when ALT >3× ULN; bleeding risk higher • HF/shock: avoid β-blocker in acute HF or shock; add Spironolactone if EF <40% with DM or HF; ACE inhibitor/ARNI for EF <40% • GI (PUD/GERD): add PPI with DAPT in patients at GI-bleeding risk (Pantoprazole; avoid Omeprazole with Clopidogrel) • LDL-C goal <55 mg/dL (add Ezetimibe if ≥70 on maximal statin, 2025 ACC/AHA); Counseling: DAPT adherence for 12 months (never stop early), bleeding signs, Ticagrelor dyspnea and Aspirin dose ≤100 mg, Nitroglycerin use (call 911 if pain persists 5 min after first dose), statin muscle pain, cardiac rehab, return to activity

Case 7: Anticoagulation (INR 4.5 with Infection)

Case details: Patient on Warfarin with an infection; INR 4.5.

Question	Answer
Q1. What to do?	• Cause: antibiotic interaction (eg TMP-SMX, metronidazole, FQ, azoles) + fever/infection increases sensitivity • No bleeding, INR 4.5: hold 1–2 doses; check INR daily–every few days • Resume at a 5–20% lower weekly dose while on the antibiotic • Recheck INR after the antibiotic course ends
Also be ready for (exam add-ons)	• Goals: keep INR in range (2–3 for most indications; 2.5–3.5 mechanical mitral valve), stop bleeding, avoid thrombosis • Non-pharm: local pressure (10–15 min) for nasal bleeding, identify the cause (dose, interacting drug, infection, diet, alcohol), consistent Vitamin K intake, written dose calendar • First-line: INR >4.5–10 without bleeding: hold 1–2 doses, reduce weekly dose 10–20%; no routine Vitamin K unless high bleeding risk (Vitamin K 1–2.5 mg PO); major bleeding: Vitamin K 5–10 mg IV + 4-factor PCC • Pregnancy: Warfarin is teratogenic (6–12 weeks) and crosses the placenta: switch to LMWH before week 6 (mechanical valves: specialist decision); Warfarin is safe during breastfeeding • Renal/dialysis: higher bleeding risk, lower dose requirement; consider Apixaban in AF with dialysis (non-valvular) • Liver: baseline INR high, lower doses, closer monitoring • HF: congestion raises INR sensitivity; interacting drugs: Amiodarone (30–50% dose cut), Fluconazole, Metronidazole, TMP-SMX, quinolones, Rifampin (increase dose), Carbamazepine • Counseling: bleeding signs, steady greens, avoid alcohol binges and NSAIDs/Aspirin, report new drugs/antibiotics, never double a missed dose, INR 3–5 days after any change, carry a card

Case 8: Cardiac Counseling (Multiple Drugs)

Case details: Patient on Amiodarone, Warfarin, Digoxin, Simvastatin, Carvedilol. Counseling station observed by camera.

Question	Answer
Q1. Provide counseling; check interactions and contraindications.	• Open: introduce yourself, confirm identity, ask what the patient knows • Amiodarone + Warfarin: Warfarin dose ↓30–50%; INR weekly • Amiodarone + Digoxin: Digoxin dose ↓30–50%; level 0.5–0.9 ng/mL • Amiodarone + Simvastatin: max 20 mg or switch statin • Amiodarone/Digoxin + Carvedilol: bradycardia, AV block • Amiodarone: with food, sunscreen, report cough/dyspnea, vision or thyroid symptoms • Warfarin: same time daily, consistent Vitamin K, no NSAIDs/herbals, bleeding signs • Digoxin: toxicity signs (nausea, visual changes, slow pulse) • Simvastatin: report muscle pain; avoid grapefruit • Carvedilol: with food, do not stop abruptly, rise slowly • Close: teach-back
Also be ready for (exam add-ons)	• Goals: confirm understanding, prevent drug interactions and adverse effects, improve adherence • Non-pharm: introduce yourself, use open questions and teach-back, written schedule, pill organizer, diet (grapefruit, Vitamin K, alcohol), sun protection • Key points by drug: Amiodarone (photosensitivity, thyroid/lung/liver, interactions), Warfarin (INR, bleeding, Vitamin K), Digoxin (nausea, vision changes, pulse <60), Simvastatin (max 20 mg with Amiodarone, muscle pain), Carvedilol (take with food, dizziness, do not stop abruptly) • Interactions: Amiodarone → ↓ Warfarin dose 30–50%, ↓ Digoxin 50%, Simvastatin ≤20 mg; Carvedilol + Amiodarone/Digoxin: bradycardia/AV block; Carvedilol hides hypoglycemia • Pregnancy: Warfarin, Amiodarone, statins contraindicated; plan contraception or switch • Renal/dialysis: Digoxin reduce dose; Warfarin sensitivity; Liver: statin and Amiodarone monitoring • Monitoring: INR, HR/ECG, digoxin level (0.5–0.9 ng/mL), TSH/LFTs every 6 months, CXR/PFT yearly (Amiodarone), eye exam, LDL

Case 9: Sinus Bradycardia

Case details: Symptomatic sinus bradycardia with hypotension not responding to Atropine.

Question	Answer
Q1. What will you give, and what is the dose?	• Atropine: 1 mg IV q3–5 min, max 3 mg (already tried) • Next: transcutaneous pacing and/or ◦ Dopamine **5–20 mcg/kg/min** ◦ or Epinephrine 2–10 mcg/min • Then: transvenous pacing • Treat the cause: beta-blocker/CCB toxicity (glucagon, calcium), Digoxin toxicity (Fab)
Also be ready for (exam add-ons)	• Goals: restore heart rate and perfusion, find the cause (ischemia, drugs, electrolytes, hypothyroidism) • Non-pharm: monitor, oxygen only if hypoxic, IV access, 12-lead ECG, transcutaneous pacing if unstable, review AV nodal blockers • First-line: Atropine 1 mg IV q3–5 min (max 3 mg); if no response: Dopamine 5–20 mcg/kg/min (or Epinephrine 2–10 mcg/min) or pacing • Pregnancy: same ACLS algorithm; left lateral position; Atropine acceptable • Renal/dialysis: check K; Liver: none; HF/β-blocker toxicity: Glucagon 3–5 mg IV; Digoxin toxicity: Digoxin immune fab • Counseling: avoid combinations of AV nodal blockers, report dizziness/syncope

2015

Case 1: NSTEMI (Oral)

Case details: Oral NSTEMI case.

Question	Answer
Q1. Causes of ischemia?	• Plaque rupture with thrombus (most common) • Supply–demand mismatch: anemia, tachyarrhythmia, hypotension, hypoxemia, thyrotoxicosis • Others: vasospasm, coronary dissection
Q2. AF and dilated cardiomyopathy: relevance?	• AF: worsens ischemia (rate); needs rate control and OAC per CHA2DS2-VASc • Dilated cardiomyopathy: LV dysfunction → GDMT
Q3. Typical vs atypical symptoms?	• Typical: retrosternal pressure radiating to arm/jaw, with exertion, sweating • Atypical (women, elderly, DM): dyspnea, fatigue, epigastric pain, nausea

Question	Answer
Q4. What labs to order?	• Serial troponin • 12-lead ECG • CBC • BMP/SCr • Lipid panel • A1C • TSH • PT/aPTT
Q5. Acute management, with regimen?	• Aspirin 162–325 mg • P2Y12: Ticagrelor 180 mg then 90 mg BID • Anticoagulant: Enoxaparin 1 mg/kg SC q12h (q24h if CrCl <30) or UFH • Nitroglycerin 0.4 mg SL q5min × 3, IV 10 mcg/min if persistent • Metoprolol tartrate 25–50 mg PO q6–12h within 24 h (no HF/shock) • Atorvastatin 80 mg daily • Invasive strategy timing by risk • Monitor: ECG, troponin, BP, HR, Hgb/platelets, SCr, bleeding
Q6. Long-term management?	• DAPT ≥12 months • Beta-blocker • ACE inhibitor/ARB • High-intensity statin ± Ezetimibe • SL Nitroglycerin PRN • Cardiac rehabilitation • Doses: ◦ Aspirin 81 mg daily (indefinitely) ◦ Ticagrelor 90 mg BID or Clopidogrel 75 mg daily for 12 months ◦ Atorvastatin 80 mg daily ± Ezetimibe 10 mg ◦ Metoprolol succinate 25 → 200 mg daily or Bisoprolol 1.25 → 10 mg daily ◦ Lisinopril 2.5–5 → 20–40 mg daily ◦ Nitroglycerin 0.4 mg SL q5min × 3 PRN chest pain • Monitor: chest pain, bleeding (Hgb), BP, HR, SCr/K (ACE inhibitor), lipids 4–8 weeks, LFT baseline, adherence to DAPT • Non-pharm: smoking cessation, diet, exercise, weight control
Also be ready for (exam add-ons)	• Goals: relieve ischemia, prevent MI and death, treat HF and hypertension, secondary prevention • Non-pharm: early invasive strategy by risk (GRACE), cardiac rehab, smoking cessation, diet (Mediterranean, sodium restriction), exercise, weight control, vaccines • First-line: Aspirin 162–325 mg, P2Y12 inhibitor (Ticagrelor 180 mg), Enoxaparin 1 mg/kg q12h or UFH, oral β-blocker early if reduced LVEF or HF without shock (reassess at 1 year if LVEF preserved), Nitroglycerin 5–10 mcg/min for persistent ischemia, hypertension or pulmonary edema; Atorvastatin 80 mg • Pregnancy: ACE inhibitor/ARB/MRA/statin contraindicated; Metoprolol, Labetalol and Nifedipine are used; low-dose Aspirin and Clopidogrel are acceptable; amiodarone avoided • Renal/dialysis: avoid Sodium nitroprusside (cyanide/thiocyanate); Enoxaparin 1 mg/kg q24h if CrCl <30; contrast prevention; loop diuretic higher doses • Liver: statin and Amiodarone hepatotoxicity monitoring; Carvedilol avoid in severe liver disease • HF (ADHF): Furosemide 20–40 mg IV (or 2.5× the home dose); avoid non-DHP CCB and Flecainide/Propafenone/Sotalol/Dronedarone; Amiodarone or Dofetilide for AF • GI (PUD/GERD): PPI with DAPT or anticoagulant; avoid NSAIDs • Counseling: DAPT 12 months, statin adherence (stop and call for severe muscle pain), Nitroglycerin use, daily weights, BP log, warning signs of MI/HF

Case 2: Cardiac Long Case ★ Repeated case, also in 2024

Case details: NSTEMI or unstable angina with HF, new-onset AF, structural heart disease, dyslipidemia, pulmonary edema, DM and HTN.

Question	Answer
Q1. Management plans, monitoring parameters and cut-off for stopping statins?	• ACS: antiplatelets, anticoagulant, nitrate, statin; beta-blocker once euvolemic • Pulmonary edema: IV loop diuretic + IV Nitroglycerin • HFrEF: GDMT once stable • AF: rate control (beta-blocker/Digoxin; avoid non-DHP CCB), Amiodarone if needed; OAC by CHA2DS2-VASc • DM/HTN: BP <130/80, SGLT2i/GLP-1 RA • Monitoring: SCr, K, Mg, INR, CBC, ECG/QTc, weight, UOP • Stop statin: CK >10× ULN with symptoms or rhabdomyolysis; persistent ALT >3× ULN • Key doses: ◦ Furosemide IV 1–2.5× home oral dose; Nitroglycerin IV 10 mcg/min ◦ Aspirin 162–325 mg then 81 mg; Ticagrelor 180 mg then 90 mg BID; Enoxaparin 1 mg/kg q12h ◦ Atorvastatin 80 mg; Apixaban 5 mg BID for AF

Question	Answer
Also be ready for (exam add-ons)	• Goals: relieve ischemia, prevent MI and death, treat HF and hypertension, secondary prevention • Non-pharm: early invasive strategy by risk (GRACE), cardiac rehab, smoking cessation, diet (Mediterranean, sodium restriction), exercise, weight control, vaccines • First-line: Aspirin 162–325 mg, P2Y12 inhibitor (Ticagrelor 180 mg), Enoxaparin 1 mg/kg q12h or UFH, oral β-blocker early if reduced LVEF or HF without shock (reassess at 1 year if LVEF preserved), Nitroglycerin 5–10 mcg/min for persistent ischemia, hypertension or pulmonary edema; Atorvastatin 80 mg • Pregnancy: ACE inhibitor/ARB/MRA/statin contraindicated; Metoprolol, Labetalol and Nifedipine are used; low-dose Aspirin and Clopidogrel are acceptable; amiodarone avoided • Renal/dialysis: avoid Sodium nitroprusside (cyanide/thiocyanate); Enoxaparin 1 mg/kg q24h if CrCl <30; contrast prevention; loop diuretic higher doses • Liver: statin and Amiodarone hepatotoxicity monitoring; Carvedilol avoid in severe liver disease • HF (ADHF): Furosemide 20–40 mg IV (or 2.5× the home dose); avoid non-DHP CCB and Flecainide/Propafenone/Sotalol/Dronedarone; Amiodarone or Dofetilide for AF • GI (PUD/GERD): PPI with DAPT or anticoagulant; avoid NSAIDs • Counseling: DAPT 12 months, statin adherence (stop and call for severe muscle pain), Nitroglycerin use, daily weights, BP log, warning signs of MI/HF